

BanglaFactGuard

A Multimodal Bangla Fact-Checking System

Final Report
SE 801: Software Project Lab III

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Letter of Transmittal

11 October 2026

BSSE 4th Year Exam Committee

Institute of Information Technology

University of Dhaka

Subject: Final Report Submission, BanglaFactGuard: A Multimodal Bangla Fact Checking System.

Dear Sir,

I am pleased to submit the final report for my Software Project Lab III project, BanglaFactGuard: A Multimodal Bangla Fact-Checking System. The report covers the project from the requirements and the analysis models to the component-level design, the interface design, the implementation and the testing of the finished system. It closes with a user manual, installation steps and the address of the public repository.

I have tried my best to present the technical details of the project clearly and accurately. However, the work may still lack perfection in places. I would therefore be grateful if you would kindly accept my report and provide any feedback that could help me improve it further.

Thank you for your consideration.

Sincerely,

Md. Abul Kalam Azad

Roll: BSSE-1425

Institute of Information Technology

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Supervisor's Signature

Toukir Ahammed

Acknowledgement

I am sincerely grateful to the Institute of Information Technology, University of Dhaka, for giving me the opportunity to carry out my Software Project Lab III on the project titled “BanglaFactGuard: A Multimodal Bangla Fact-Checking System”. This project gave me the chance to apply software engineering, natural language processing, computer vision and information retrieval to a practical and socially relevant problem.

I would like to express my heartfelt gratitude to my respected supervisor, Toukir Ahammed, Lecturer, Institute of Information Technology, University of Dhaka, for his continuous guidance, constructive criticism and encouragement throughout every phase of this work. His insights shaped the direction, design and scope of this system.

I am also thankful to my other teachers and classmates for their cooperation, helpful discussions and suggestions during different phases of the project. Finally, I acknowledge the resources and academic environment provided by the Institute of Information Technology, which made the progress of this project possible.

Abstract

BanglaFactGuard is an AI-assisted platform that helps people check Bangla news claims, changed headlines and news photocards. It has three parts: an Angular website, a FastAPI backend and a Chrome browser extension. Users can check a claim in three ways. Text & source verification looks for the story on the news outlet that the claim names and reports three separate findings: whether a matching report was found, whether the headline was altered and whether the publication date matches. When article text is given, it also shows four body-similarity scores. Photo card verification reads the headline, outlet and printed date from an uploaded card with Gemini and then runs the same source check. Text & image verification uses a BanglaBERT and EfficientNet-B4 model to give a preliminary Likely Real or Likely Fake prediction.

All automated findings are shown as preliminary. Human experts then review each claim and give a final verdict (Fake, Real, Misleading or Altered) through credibility-weighted voting; claims without agreement are escalated to an administrator. Missing or incomplete evidence is shown honestly and is never treated as proof that a claim is false. The platform also offers a public Fact Explorer, submission history, in-app and email notifications and administration of experts, news sources and voting rules. PostgreSQL, Redis and MinIO store data, cached results and images and a durable job queue runs long checks in the background.

This report presents the requirements, activity diagrams, data and class models, AI engineering design, component-level design, interface design, implementation, testing, user manual and installation of the system.

Table of Contents

Letter of Transmittal	2
Acknowledgement	3
Abstract	4
List of Figures	8
List of Tables	10
1. Introduction.....	11
1.1 Problem Statement	11
1.2 Objectives.....	11
1.3 Scope	12
1.3.1 In Scope.....	12
1.3.2 Out of Scope.....	12
1.4 Deliverables.....	12
1.5 Functional and Non-Functional Requirements	13
1.5.1 Functional Requirements.....	13
1.5.2 Non-Functional Requirements	14
2. Activity Diagrams	15
2.1 Authentication and Registration.....	15
2.2 Text & Source Verification.....	16
2.3 Photo Card Verification	17
2.4 Text & Image (Multimodal) Verification	18
2.5 Expert Review and Voting.....	19
2.6 Browser Extension	20
2.7 Fact Explorer.....	21
3. Data-Based Modeling and Dataset Description.....	22
3.1 Entity-Relationship Diagram	22
3.2 Main Entities and Relationships.....	22
3.3 Dataset Description	24
4. Class-Based Modeling	26
5. AI Engineering Design	28
5.1 Multimodal Verification Model	28
5.1.1 Model Architecture	28
5.1.2 Recorded Evaluation	29
5.1.3 Duplicate Detection.....	Error! Bookmark not defined.
5.2 Source-Based Verification Pipeline	30
5.3 Model Selection	31
5.4 Responsible Use of AI	31
6. Component-Level Design	33

6.1 Component Diagram	33
6.1.1 Information Flow.....	34
6.2 Design Classes	35
6.3 Persistent Data Sources	37
6.4 Behavioural Design.....	39
6.4.1 Submission Lifecycle	39
6.4.2 Sequence of a Text & Source Check.....	40
6.4.3 Voting Rules.....	40
6.5 Deployment Design.....	42
6.6 Design Alternatives Considered.....	43
7. Interface Design	44
7.1 User Interface Objects and Actions.....	44
7.2 Navigation	45
7.3 UI Events and State Transitions	46
7.3.1 Verify Facts Page	46
7.3.2 Expert and Administrator Review	47
7.3.3 Browser Extension Side Panel	48
7.4 Interface States as Seen by the User	48
8. Implementation	55
8.1 Technology Stack.....	55
8.2 Code Organisation.....	55
8.3 Key Implementation Details	55
8.3.1 Authentication and Roles	55
8.3.2 Background Jobs and Reuse.....	55
8.3.3 Photo Card Reading	56
8.3.4 Voting, Escalation and Notifications	56
8.3.5 Browser Extension	56
8.4 Main API Routes.....	56
9. Testing.....	57
9.1 Testing Approach and Strategy.....	57
9.2 Test Environment and Tools	57
9.3 Item Pass/Fail Criteria.....	57
9.4 Summary of Executed Test Runs	58
9.5 Test Cases with Detailed Outcomes.....	58
9.5.1 Authentication and Sessions.....	58
9.5.2 Text & Source Verification	59
9.5.3 Photo Card and Text & Image Verification	60
9.5.4 Expert Review, Voting and Escalation	60

9.5.5 Notifications, Fact Explorer and Extension	61
9.6 Observed Performance and Model Results	61
9.7 Tests Not Executed or Not Verified	61
9.8 Risks and Contingencies	62
10. User Manual	63
10.1 Getting Started	63
10.2 Creating an Account and Signing In	63
10.3 Checking a Claim	64
10.3.1 Text & Source (News story against claimed source)	64
10.3.2 Photo Card (Photocard against claimed source)	64
10.3.3 Text & Image (News story against an image)	65
10.4 Reading a Result	66
10.5 Fact Explorer	69
10.6 My Submissions, Notifications and Settings	69
10.7 Using the Browser Extension	71
10.8 Expert Tasks	71
10.9 Administrator Tasks	73
10.10 Troubleshooting	77
11. Repository Installation and Deployment	78
11.1 Repository	78
11.2 Prerequisites	78
11.3 Backend Setup	78
11.4 Services, Database and Sources	79
11.5 Website	79
11.6 Browser Extension (Installation)	79
11.7 Running the Tests	80
11.8 Deployment Status	80
12. Conclusion	81
13. References	82

List of Figures

Figure 2.1: Activity Diagram — Authentication and Registration	15
Figure 2.2: Activity Diagram — Text & Source Verification	16
Figure 2.3: Activity Diagram — Photo Card Verification	17
Figure 2.4: Activity Diagram — Text & Image Verification	18
Figure 2.5: Activity Diagram — Expert Review and Voting	19
Figure 2.6: Activity Diagram — Browser Extension	20
Figure 2.7: Activity Diagram — Fact Explorer	21
Figure 3.1: Entity-Relationship Diagram — BanglaFactGuard	22
Figure 4.1: Analysis Class Diagram (CRC View)	26
Figure 5.1: Multimodal Verification Architecture — Text and Image Fusion	28
Figure 5.2: Source-Based Verification Pipeline — Stages S01 to S13	30
Figure 6.1: Component Diagram of BanglaFactGuard	33
Figure 6.2: Design Class Diagram	35
Figure 6.3: Persistent Data Sources and Their Classes	37
Figure 6.4: State Diagram of a Submission	39
Figure 6.5: Sequence Diagram — Text & Source Claim (rotated to fit the page)	41
Figure 6.6: Deployment Diagram	42
Figure 7.1: Website Navigation by User Role	45
Figure 7.2: State Diagram of the Verify Facts Interface	46
Figure 7.3: State Diagram of the Review Interface	47
Figure 7.4: State Diagram of the Extension Side Panel	48
Figure 7.5: EDITING State — Text & Source Claim Form	49
Figure 7.6: IN PROGRESS State — Claim Received	50
Figure 7.7: PRELIMINARY RESULT State — Final Decision Under Review	51
Figure 7.8: FINAL DECISION State — Administrator Decision on a Photo Card	52
Figure 7.9: CHECK FAILED State — Photo Card Could Not Be Read	53
Figure 7.10: Extension States — Signed In, Checking Your Claim and Result Ready	53
Figure 7.11: VOTE FORM State — Expert Review Page	54
Figure 10.1: Home Page	63
Figure 10.2: Create Account Page	64
Figure 10.3: Photo Card Form with Preview	65
Figure 10.4: Text & Image Result with a Reused Assessment	66
Figure 10.5: Headline Comparison Showing an Altered Headline	67
Figure 10.6: Body Similarity Scores	67
Figure 10.7: Result When the Claimed Source Has No Matching Report	68
Figure 10.8: Fact Explorer	69
Figure 10.9: My Submissions	70
Figure 10.10: Notifications	70
Figure 10.11: Extension Verify Tab — Text & Source, Photo Card and Text & Image	71
Figure 10.12: Expert Review Queue	72
Figure 10.13: Administrator Dashboard	73
Figure 10.14: Expert Management	74
Figure 10.15: Verified Sources	74
Figure 10.16: Voting Configuration and Credibility Weight Tiers	75
Figure 10.17: Final Decision with Voting Details After an Administrator Decision	76

List of Tables

Table 1.1: Functional Requirements	13
Table 1.2: Non-Functional Requirements	14
Table 3.1: Main Entities and Their Key Attributes	23
Table 3.2: Dataset Split Summary (Text–Image Pairs)	25
Table 4.1: Class Responsibilities and Collaborators	27
Table 5.1: Multimodal Model Configuration	29
Table 5.2: Recorded Test Results of the Multimodal Model (960 test items)	29
Table 5.3: Source-Based Verification Pipeline — Stage Reference	30
Table 5.4: AI Model Selection Summary	31
Table 6.1: Main Components and Their Responsibilities	33
Table 6.2: Design Classes and Domain Concepts	36
Table 6.3: Persistent Data Sources	38
Table 6.4: Submission States	40
Table 6.5: Deployment Nodes	42
Table 6.6: Design Alternatives and Decisions	43
Table 7.1: User Interface Objects and Actions	44
Table 7.2: Events and State Transitions on the Verify Facts Page	47
Table 8.1: Technology Stack	55
Table 8.2: Main API Routes (prefix /api/v1)	56
Table 9.1: Test Environment	57
Table 9.2: Executed Automated Test Runs (8 October 2026)	58
Table 9.3: Test Cases — Authentication and Sessions	58
Table 9.4: Test Cases — Text & Source Verification	59
Table 9.5: Test Cases — Photo Card and Text & Image	60
Table 9.6: Test Cases — Expert Review and Voting	60
Table 9.7: Test Cases — Notifications, Fact Explorer and Extension	61
Table 9.8: Tests Not Executed in This Run	61
Table 9.9: Testing Risks and Contingencies	62
Table 10.1: Common Problems and Solutions	77
Table 11.1: Prerequisites	78
Table 11.2: Main Configuration Settings (backend/.env)	79

Project Title: BanglaFactGuard: A Multimodal Bangla Fact-Checking System

GitHub Repository: <https://github.com/abulkalamazad1425/BanglaFactGuard>

1. Introduction

Fake and misleading news spreads very quickly on social media. In Bangladesh, much of this content is shared in Bangla, often as a changed headline or as a “photocard”, which is an image that looks like a news outlet's post. Most automated fact-checking tools are built for English, so Bangla readers have few tools that can check such claims quickly.

BanglaFactGuard is a full-stack fact-checking platform for Bangla news. A user submits a claim on the website or from the browser extension. The system first produces quick automated findings, which are always labelled as preliminary. Human experts then examine the claim and its evidence and vote on a final verdict. Every reviewed claim, with its evidence and verdict, can be searched in the public Fact Explorer.

1.1 Problem Statement

At present, Bangla-speaking users do not have a single place where they can check a news claim, a headline or a photocard. Manual fact-checking by journalists is accurate but slow and it cannot handle the volume of claims shared every day. Automated models are fast, but their output is not reliable enough to be the final word and a missing search result can easily be misread as proof that a story is fake.

The problem this project addresses is therefore to build one platform that (1) accepts Bangla claims in the forms in which they are actually shared, (2) gives fast, clearly preliminary evidence-based findings, (3) keeps the final decision with human experts and (4) makes the results easy to find and reuse.

1.2 Objectives

The objectives of BanglaFactGuard are:

- To let users check Bangla news claims in three ways: Text & source, Photo card and Text & image.
- To compare a claim with the report published by the claimed news outlet and give separate findings for source, headline and publication date.
- To give a quick preliminary prediction for a news text and image pair using a multimodal model.
- To keep every final verdict with human experts through credibility-weighted voting, with administrator decisions for escalated claims.
- To reuse saved results for claims that were already checked, instead of repeating the work.
- To provide a public Fact Explorer, personal submission history and result notifications.
- To provide a browser extension so users can check a claim from the page they are reading.
- To give administrators tools to manage experts, verified news sources and voting rules.

1.3 Scope

1.3.1 In Scope

- A website with four kinds of users: anonymous visitor, registered user, expert and administrator.
- Three verification methods: Text & source, Photo card (read by Gemini) and Text & image (BanglaBERT + EfficientNet-B4).
- Separate automated findings for source correspondence, headline alteration and date, plus four body-similarity scores when a body is given.
- Background processing through a durable job queue, with reuse of saved results.
- Expert review, credibility-weighted voting, escalation and administrator decisions.
- Public Fact Explorer with search, filters and statistics.
- In-app notifications for signed-in submitters and queued final-result emails.
- A Chrome (Manifest V3) extension with a side panel, selected-text import, image upload, screenshot-area capture, cropping, activity tracking and desktop alerts.
- Administration of experts, verified news sources, credibility tiers and voting configuration.

1.3.2 Out of Scope

- Checking claims in languages other than Bangla.
- Native mobile applications; the system is a website and a desktop Chrome extension.
- Video verification.
- Fully automated final verdicts; every final verdict comes from experts or an administrator.
- Automatic crawling of social media; only submitted claims are checked.
- Removing content or taking legal action against publishers.

1.4 Deliverables

- The BanglaFactGuard website (Angular), backend API (FastAPI) and Chrome extension.
- The trained multimodal model files used by the Text & image method.
- The database design and migrations and the verified-source seeding script.
- Automated test suites for the backend, the website and the extension.
- This final report, with design, implementation, testing, a user manual and installation steps.
- A public Git repository: <https://github.com/abulkalamazad1425/BanglaFactGuard>.

1.5 Functional and Non-Functional Requirements

1.5.1 Functional Requirements

Table 1.1 lists the functional requirements of the final system. They are taken from the earlier SRS and updated to match what is implemented.

Table 1.1: Functional Requirements

ID	Requirement	Users
FR-1	Register with name, email and password; sign in and sign out; reset a forgotten password with a time-limited emailed code (OTP).	All signed-in roles
FR-2	Submit a Text & source claim with a headline and claimed outlet and an optional body and claimed date.	All users
FR-3	Submit a Photo card image; read its headline, outlet and printed date; explain the reason when the card cannot be read.	All users
FR-4	Submit a Text & image claim (headline, article text and image) and receive a preliminary Likely Real or Likely Fake prediction.	All users
FR-5	Report source correspondence (Found, Not Found or Incomplete), headline alteration (Exact Matched, Meaning Preserved or Altered) and date (Matched or Mismatched) separately.	All users
FR-6	Show four body-similarity scores when article text is supplied.	All users
FR-7	Reuse a saved result for an identical claim and say that it was reused.	All users
FR-8	Run checks in the background and show progress; let the user leave the page safely.	All users
FR-9	Show an expert review queue; let experts inspect evidence and vote with a written justification (at least 50 characters).	Expert
FR-10	Finalize a verdict by credibility-weighted voting or escalate to an administrator when the review limits are reached without agreement.	Expert, Administrator
FR-11	Let an administrator decide escalated claims; other claims stay view-only.	Administrator
FR-12	Provide a public Fact Explorer with search, filters and statistics.	All users
FR-13	Show each signed-in user's submissions and notifications; queue final-result emails.	Registered user
FR-14	Manage expert accounts, verified sources, credibility tiers and voting configuration.	Administrator
FR-15	Offer all three methods in the browser extension, with selected-text import, screenshot capture, cropping, activity and alerts.	All users

1.5.2 Non-Functional Requirements

Table 1.2 lists the main non-functional requirements and how the final system meets them.

Table 1.2: Non-Functional Requirements

ID	Category	Requirement and how it is met
NFR-1	Performance	Long checks run in the background; the website and extension stay usable and show the current status.
NFR-2	Efficiency	Saved results are reused and search, embedding and claim data are cached in Redis.
NFR-3	Security	Passwords are hashed; access uses short-lived JWT tokens with rotating refresh tokens; OTP codes are single-use and expire.
NFR-4	Security	Role checks protect expert and admin functions; private records check ownership; experts cannot vote on their own claims.
NFR-5	Usability	Bangla input is supported; findings use plain labels and short explanations; a public FAQ explains the terms.
NFR-6	Reliability	Accepted jobs survive restarts; stalled jobs are re-claimed; failures are saved with a readable reason.
NFR-7	Data integrity	A submission and its job are saved in one transaction; persistence is idempotent; the weight of every vote is stored.
NFR-8	Correctness and transparency	Automated findings never set the final verdict; a missing source is never shown as Fake.
NFR-9	Maintainability	Features are separate modules; pipeline stages share one interface; voting rules and sources are changed without code changes.
NFR-10	Compatibility	The extension uses Chrome Manifest V3 and the same accounts as the website, with a separate login session.

2. Activity Diagrams

This chapter shows the activity diagrams for the most important workflows of BanglaFactGuard. They were drawn for the SRS and updated to match the final system. Less important flows, such as profile management, notifications and source management, are left out to save space; their behaviour is covered in Chapters 7 and 10.

2.1 Authentication and Registration

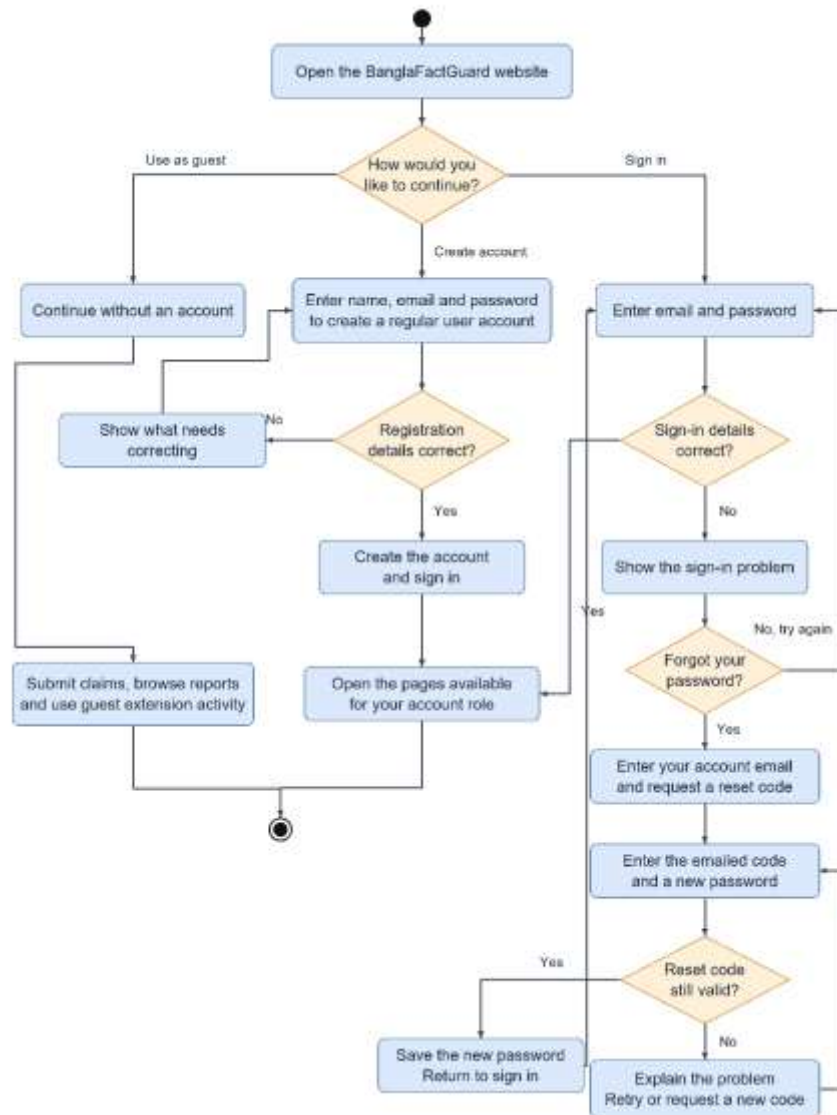


Figure 2.1: Activity Diagram — Authentication and Registration

A visitor can continue as a guest, create an account or sign in. Registration checks the details and signs the new user in. A failed sign-in shows the problem. A user who forgot the password asks for a reset code by email, enters the code with a new password and returns to sign in. After sign-in, the user sees the pages allowed for his or her role.

2.2 Text & Source Verification

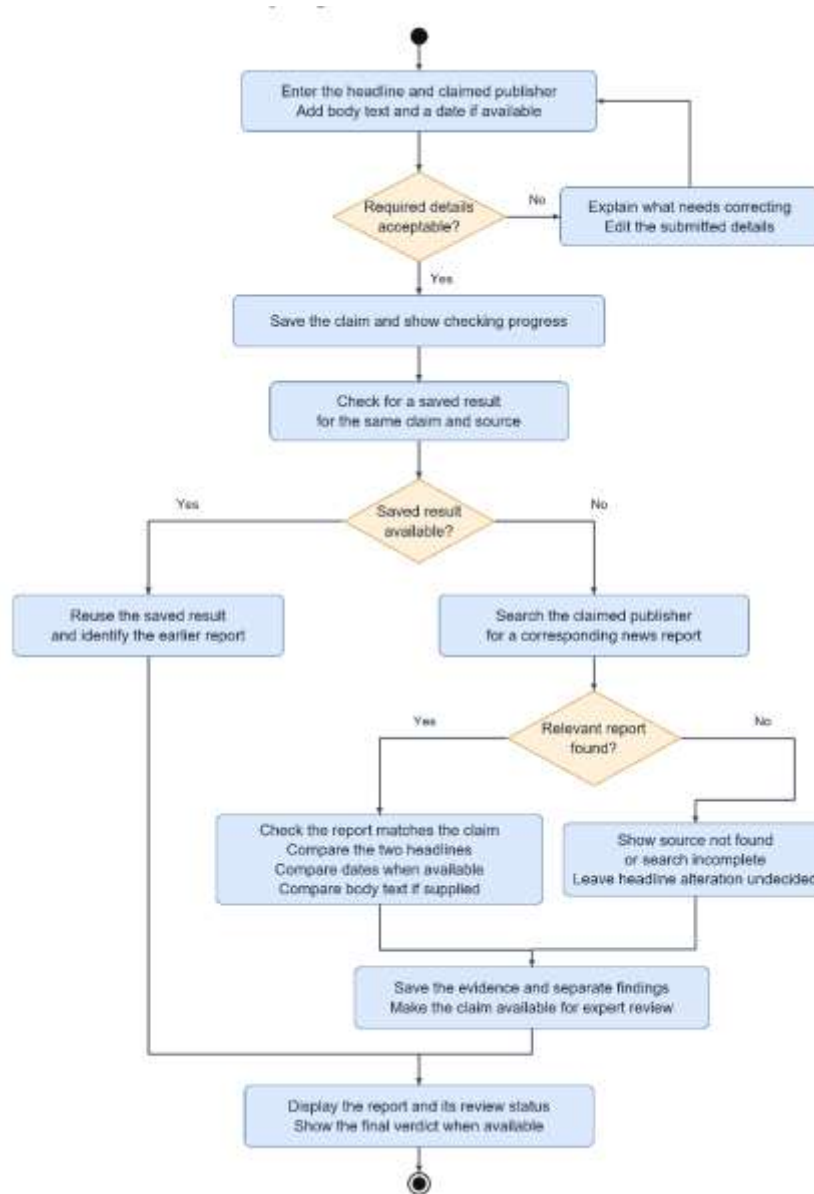


Figure 2.2: Activity Diagram — Text & Source Verification

The user enters the headline and claimed outlet and optionally the body and date. If a saved result for the same claim exists, it is reused. Otherwise the system searches the claimed outlet. When a relevant report is found, the headline, date and body are compared; when it is not found, the result says Not Found or Incomplete and the headline check stays undecided. The findings are saved and the claim goes to expert review.

2.3 Photo Card Verification

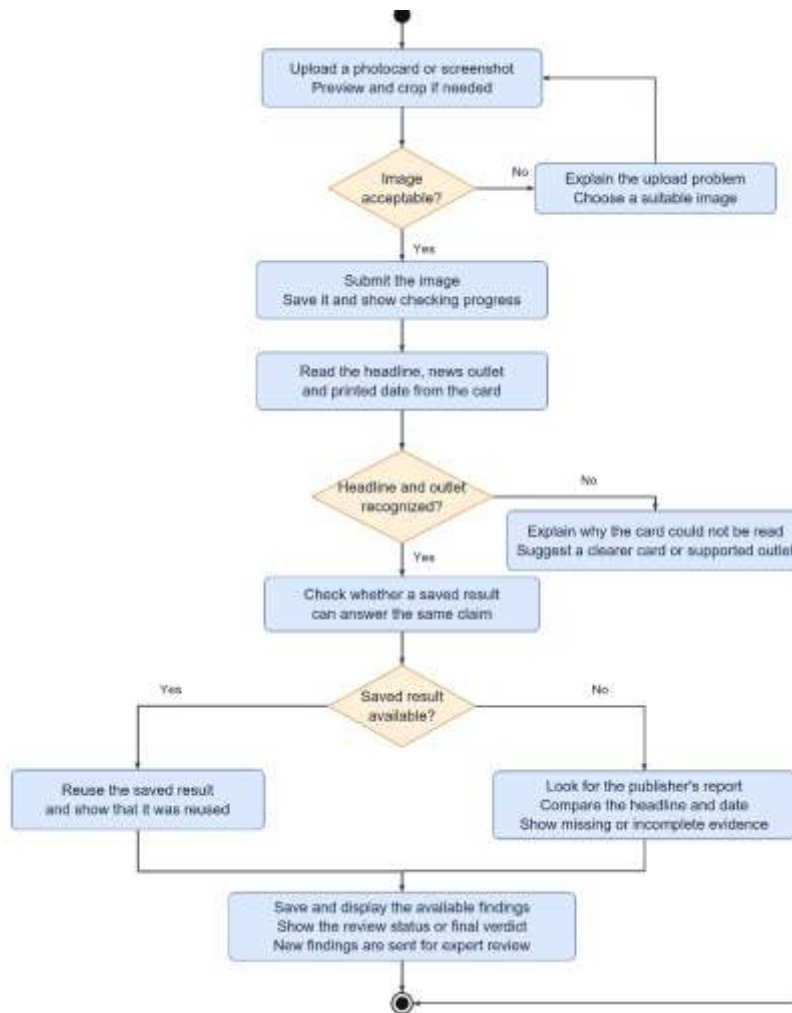


Figure 2.3: Activity Diagram — Photo Card Verification

The user uploads only the card image. The system reads the headline, the outlet and the printed date from the card. If the headline or a known outlet cannot be recognised, the user is told why. Otherwise the card is checked in the same way as a text claim and the findings go to expert review.

2.4 Text & Image (Multimodal) Verification

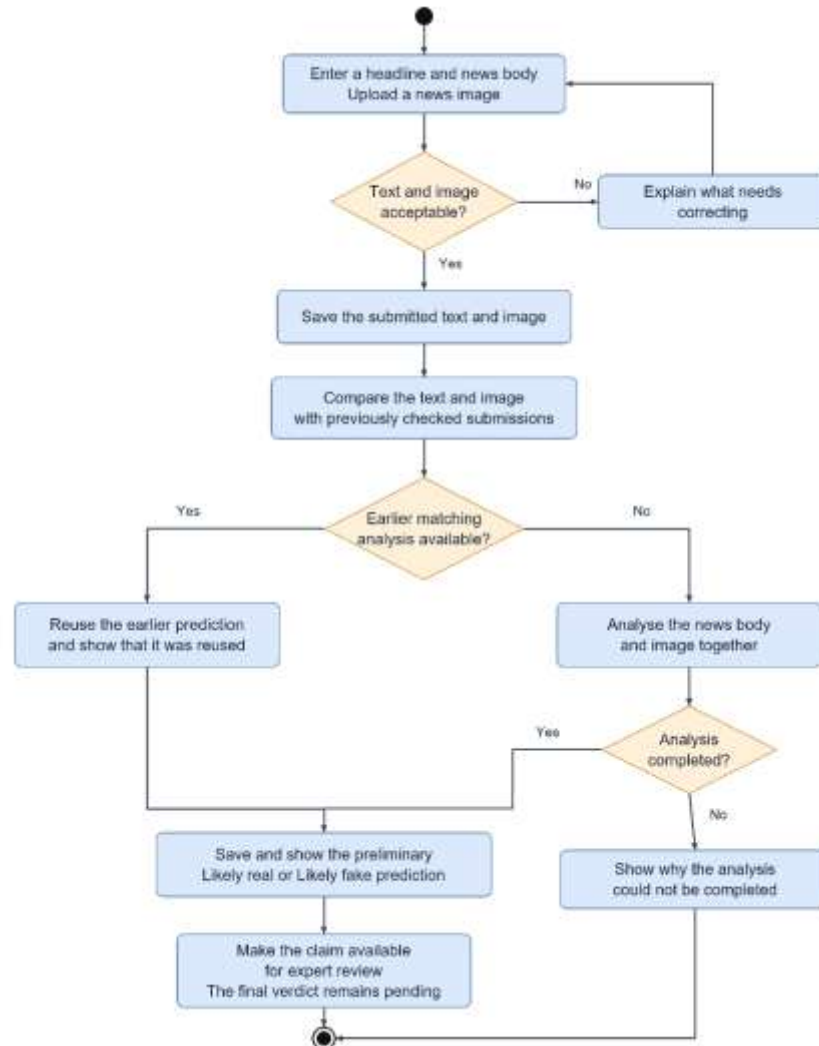


Figure 2.4: Activity Diagram — Text & Image Verification

The user enters a headline and the news text and uploads the news image. The system first compares the text and image with earlier submissions. A close match reuses the earlier prediction. Otherwise the model analyses the text and image together and saves a preliminary Likely Real or Likely Fake prediction. The claim then waits for expert review.

2.5 Expert Review and Voting

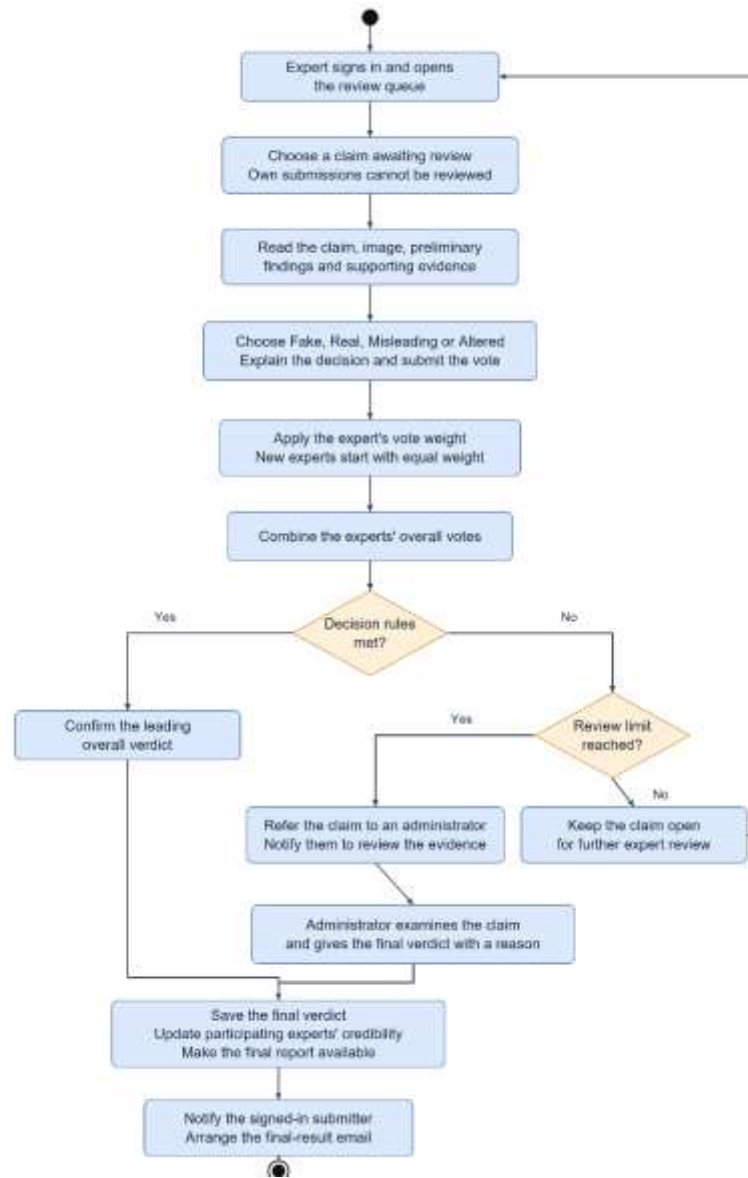


Figure 2.5: Activity Diagram — Expert Review and Voting

An expert opens the queue, chooses a claim (never his or her own), reads the evidence and votes Fake, Real, Misleading or Altered with a justification. The system applies the expert's vote weight and combines the votes. If the decision rules are met, the leading verdict becomes final. If a review limit is reached without agreement, an administrator decides. The final verdict is saved, expert credibility is updated and the submitter is notified.

2.6 Browser Extension

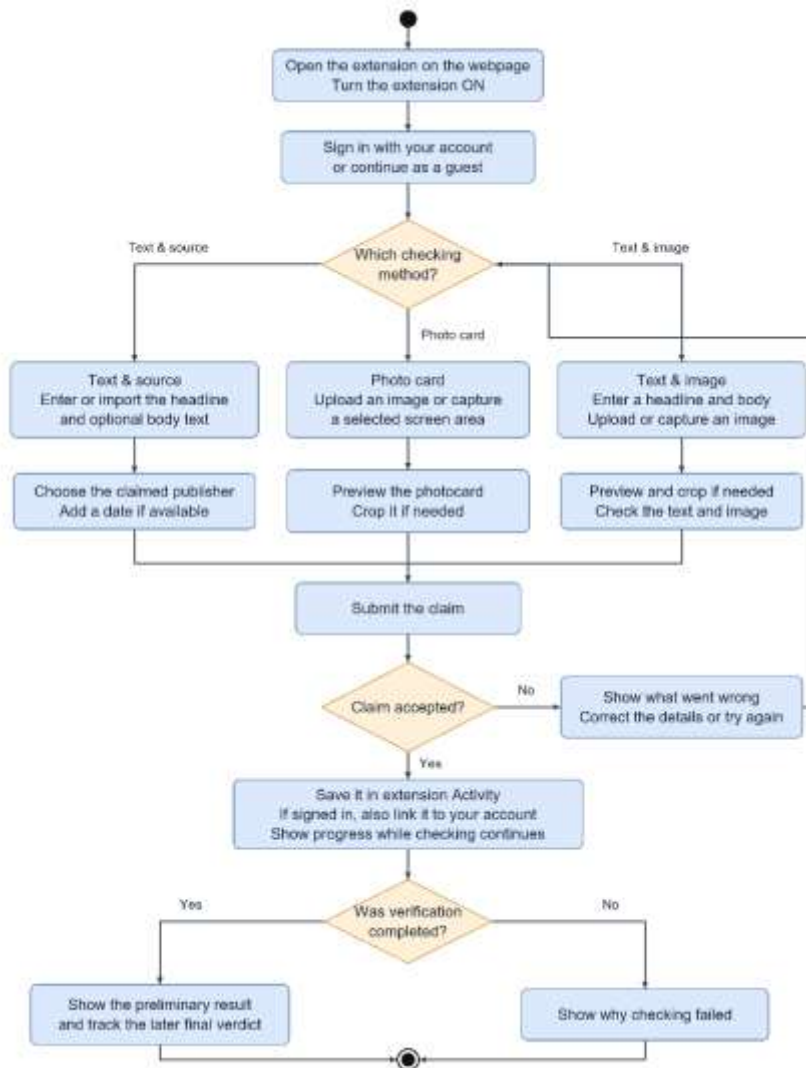


Figure 2.6: Activity Diagram — Browser Extension

The user opens the side panel, signs in or continues as a guest and chooses a method. Text can be typed or imported from the page; images can be uploaded or captured from a selected screen area and cropped. After submission the claim is saved in Activity and the panel shows progress, the preliminary result and, later, the final verdict or the reason the check failed.

2.7 Fact Explorer

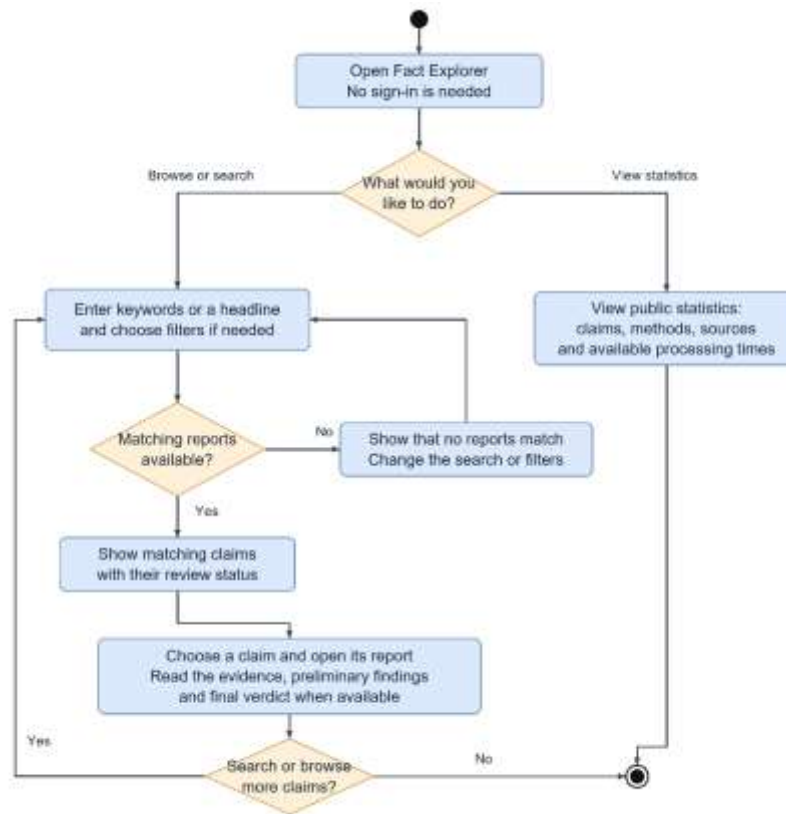


Figure 2.7: Activity Diagram — Fact Explorer

Anyone can open the Fact Explorer without signing in. The user searches by keyword or headline and applies filters or views the public statistics. Matching claims are shown with their review status and the user can open a full report to read the evidence and the final verdict when available.

3. Data-Based Modeling and Dataset Description

3.1 Entity-Relationship Diagram

The ER diagram in Figure 3.1 shows the core entities of BanglaFactGuard and how they are related. User is specialised into RegisteredUser, Expert and Administrator. Submission is the centre of the model; every check, its evidence, its result and its expert reviews belong to one submission.

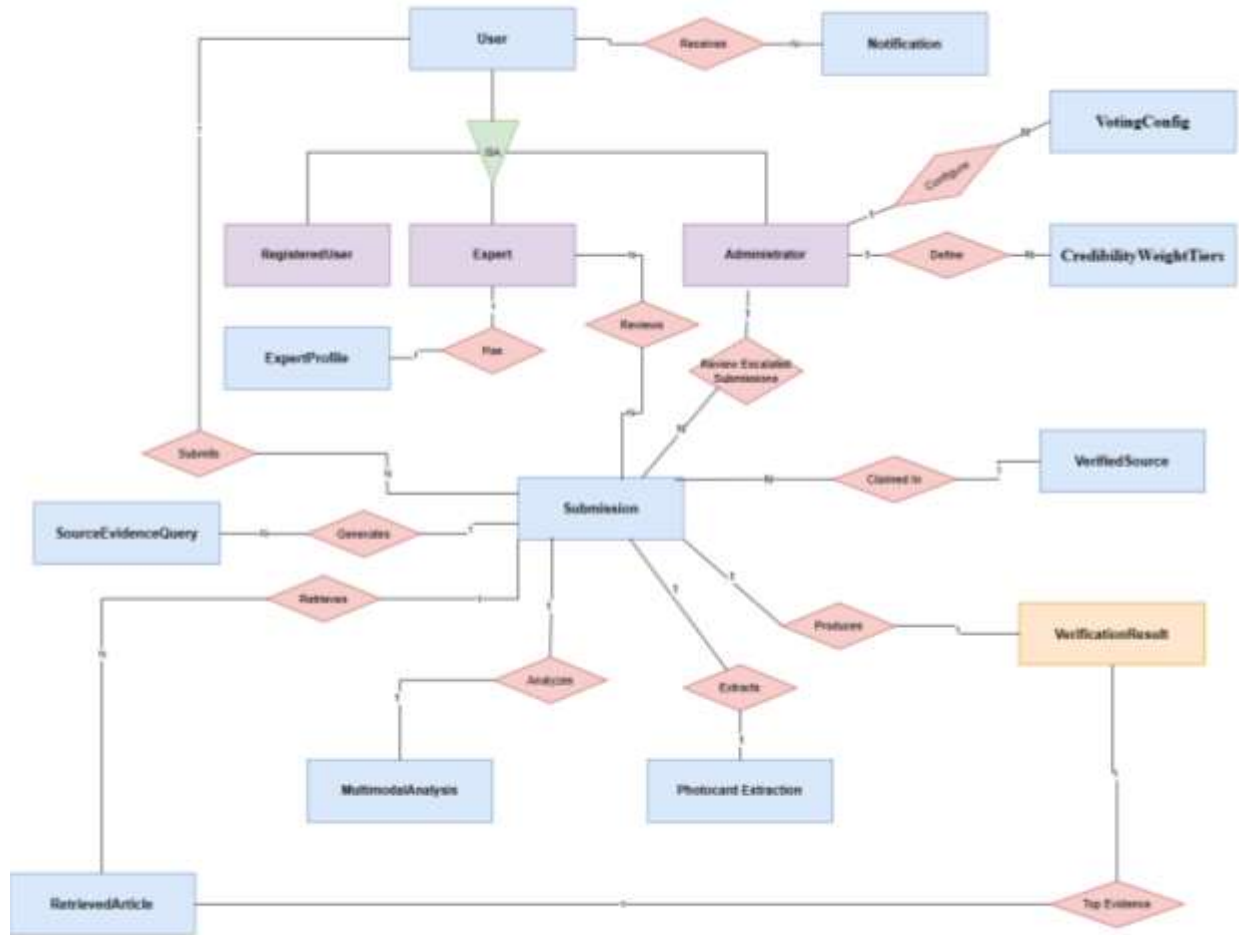


Figure 3.1: Entity-Relationship Diagram — BanglaFactGuard

3.2 Main Entities and Relationships

The main relationships are:

- A User submits many Submissions and receives many Notifications (1:N).
- An Expert has one ExpertProfile (1:1) and reviews many Submissions; a Submission is reviewed by many Experts (M:N, stored as ExpertReview records).

- An Administrator configures the VotingConfig, defines CredibilityWeightTiers and reviews escalated Submissions.
- Each Submission claims one VerifiedSource; one source can be claimed by many Submissions (N:1).
- A Submission generates many SourceEvidenceQueries and retrieves many RetrievedArticles (1:N).
- A Submission has at most one MultimodalAnalysis, one PhotocardExtraction and one VerificationResult (1:1); a VerificationResult points to one RetrievedArticle as its top evidence.

Table 3.1 summarises the main tables that implement these entities. The system also uses supporting tables for tokens, the job queue and result delivery (Section 6.3).

Table 3.1: Main Entities and Their Key Attributes

Entity (table)	Purpose	Key attributes
User (users)	Account of every signed-in person	email, hashed_password, full_name, role, is_active, total_submissions
ExpertProfile (expert_profiles)	Expert details and accuracy record	user_id, area_of_expertise, credential_notes, credibility_score, total_votes, correct_votes, completed_reviews_count, is_active
VerifiedSource (verified_sources)	Registered Bangla news outlet	canonical_name, display_name, display_name_en, aliases, base_url, rss_url, language, search_language, internal_search_url, body_selectors, title_selectors, date_selectors, article_url_patterns, js_rendered, description, is_active
Submission (submissions)	One submitted claim	submitter_id, submission_type, headline, body_text, claimed_source_text, claimed_source_id, published_date, content_hash, duplicate_of_submission_id, status, processing_phase, failure_reason, is_published, view_count, escalated_at
RetrievedArticle (retrieved_articles)	Article fetched as evidence	submission_id, url, url_hash, title, body, author, published_date, extraction_method, extraction_success, rank_score, retrieved_at
SourceEvidenceQuery (source_evidence_queries)	Search query that was run	submission_id, query_type, query_text, search_provider, results_count, executed_at

Entity (table)	Purpose	Key attributes
VerificationResult (verification_results)	Automated findings and final verdict	submission_id, source_status, content_status, date_status, headline_check_status, headline_exact_match, body_comparison_status, top_article_id, headline_similarity, headline_keyword_coverage, passage_keyword_coverage, claim_scope, pipeline_version, analysis_details, overall_verdict, finalized_at, reused_from_submission_id
PhotocardExtraction (photocard_extractions)	What Gemini read from a card	submission_id, image_object_key, status, failure_code, model_version, attempts, extraction_details
MultimodalAnalysis (multimodal_analysis)	Text & image prediction	submission_id, image_object_key, prediction, confidence_fake, confidence_real, text_embedding, image_embedding, combined_embedding, model_version, expert_overall_verdict, finalized_at,
ExpertReview (expert_reviews)	One expert or admin vote	submission_id, reviewer_id, ai_overall_verdict, ai_source_status, ai_content_status, ai_date_status, vote_overall_verdict, vote_source_status, vote_content_status, vote_date_status, justification, credibility_weight, applied_weight_tier_id, status, is_admin_decision
VotingConfig	Voting rules set by the admin	min_expert_votes, activation_threshold_votes, verified_threshold, lead_margin, max_review_votes, max_review_hours
CredibilityWeightTier	Experts Credibility weight tier	label, min_accuracy_pct, max_accuracy_pct, weight, is_active
Notification (notifications)	In-app message or Gmail	user_id, title, body, notification_type, link_url, is_read

3.3 Dataset Description

The AI model is trained on MultiBanFakeDetect, a Bangla multimodal fake-news dataset [1]. Each instance pairs a news headline and description with an image and a label: non-fake or one of three fake types (misinformation, rumor or clickbait). The news comes from online forums, news websites and social media and covers topics such as politics, sports, technology, national news, entertainment, education, international news, crime and business.

The dataset is balanced between fake and non-fake items and is split by stratified sampling into training, validation and test sets in an 80/10/10 ratio. Several trained annotators labelled each item and agreement was checked with Cohen's and Fleiss' Kappa. BanglaFactGuard uses the two-class form of the labels (Real = 0, Fake = 1).

Table 3.2: Dataset Split Summary (Text–Image Pairs)

Subset	Instances	Class balance
Training	7,680	Stratified across the label classes
Validation	960	Stratified across the label classes
Testing	960	480 Real and 480 Fake (as recorded in the evaluation run)
Total	9,600	4,800 fake and 4,800 non-fake

The Text & source and Photo card methods do not use a training dataset. They use pretrained language models and live articles from the registered Bangla news outlets (ten active outlets in the demonstration database).

4. Class-Based Modeling

Class-based modeling identifies the main analysis classes of the problem domain, their responsibilities and the classes they work with. Figure 4.1 shows these classes with their main attributes and operations and Table 4.1 gives their responsibilities and collaborators in CRC form. Chapter 6 refines these analysis classes into design classes.

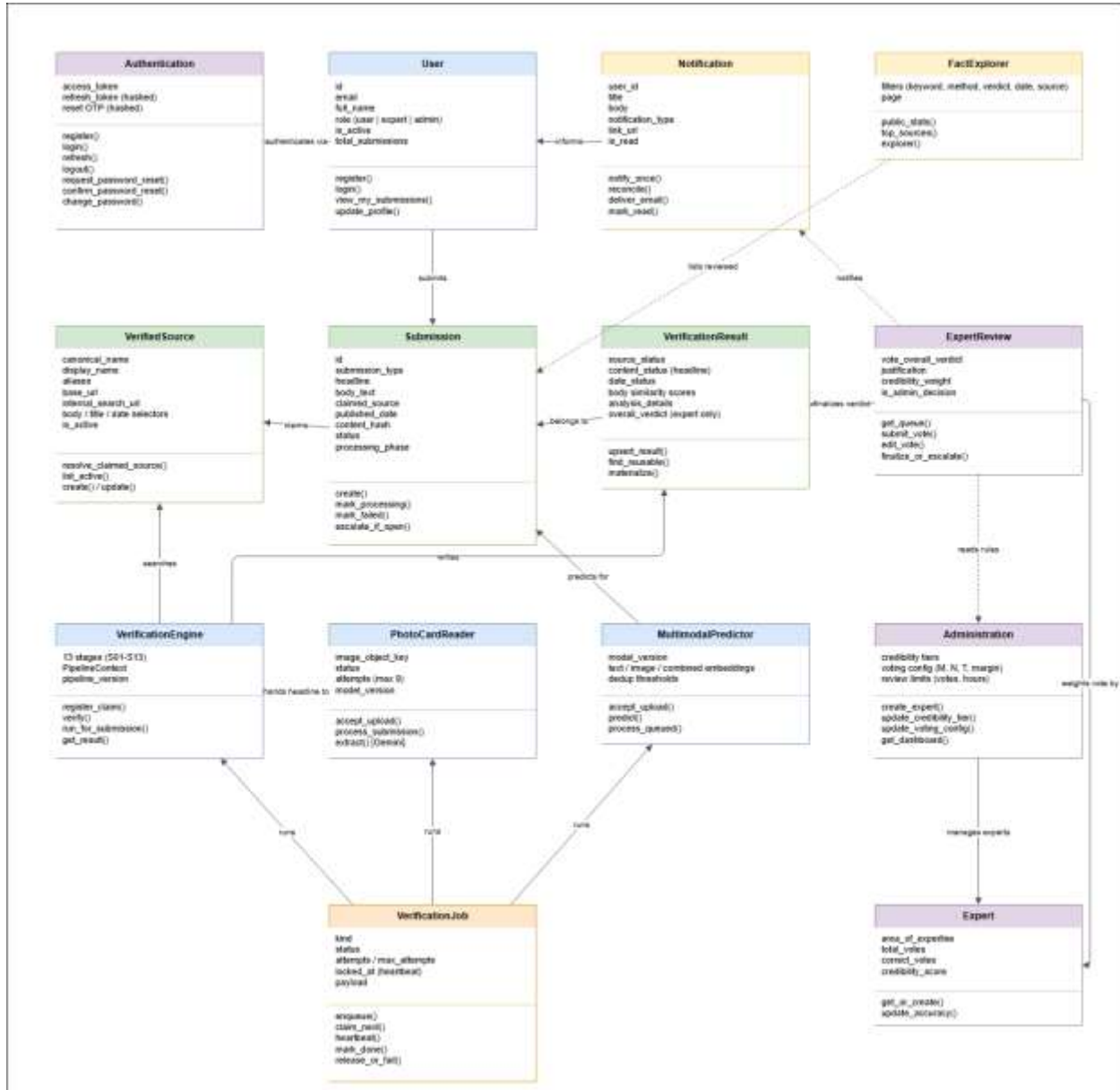


Figure 4.1: Analysis Class Diagram (CRC View)

Table 4.1: Class Responsibilities and Collaborators

Class	Responsibilities	Collaborators
User	Registers, signs in, views own submissions, updates profile	Authentication, Submission, Notification
Authentication	Issues and refreshes tokens, signs out, resets passwords with OTP	User
Submission	Holds a claim and its status and phase; marks processing, failure and escalation	VerifiedSource, VerificationResult, ExpertReview
VerifiedSource	Resolves a claimed outlet and lists active sources	Submission, VerificationEngine
VerificationEngine	Runs the 13-stage source check and returns findings	Submission, VerifiedSource, VerificationResult
PhotoCardReader	Accepts a card, reads it with Gemini and hands the headline to the engine	VerificationEngine, Submission
MultimodalPredictor	Accepts a text and image, predicts, detects duplicates	Submission
VerificationJob	Queues, claims, heartbeats, completes or fails background work	VerificationEngine, PhotoCardReader, MultimodalPredictor
VerificationResult	Stores findings; finds and copies reusable results	Submission, ExpertReview
ExpertReview	Builds the queue, records votes, finalizes or escalates	Expert, Administration, Notification
Expert	Keeps accuracy and credibility	ExpertReview, Administration
Administration	Manages experts, tiers, voting rules and the dashboard	Expert, ExpertReview
Notification	Sends each notification once, delivers email, marks read	User, ExpertReview
FactExplorer	Lists reviewed claims, statistics and top sources	Submission

5. AI Engineering Design

BanglaFactGuard uses AI in two places described in this chapter: the multimodal model behind Text & image verification and the source-based verification pipeline behind Text & source verification. Both give preliminary findings only; the final verdict always comes from human reviewers.

5.1 Multimodal Verification Model

5.1.1 Model Architecture

Figure 5.1 shows the multimodal design from the SRS. The news text is encoded by BanglaBERT and the news image by EfficientNet-B4. The two feature vectors are joined (early fusion) and passed through fully connected layers that output Real or Fake.

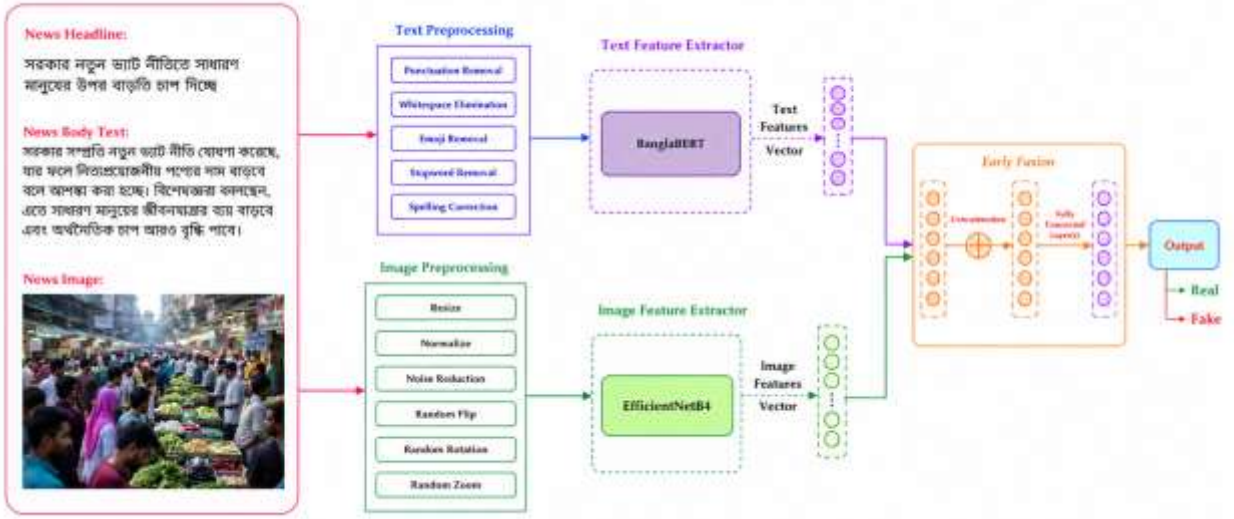


Figure 5.1: Multimodal Verification Architecture — Text and Image Fusion

In the implemented model (MultiFusionFake), BanglaBERT gives the 768-value [CLS] vector of the body text (up to 128 tokens) and EfficientNet-B4 gives a 1,792-value image vector (images resized to 380×380 and normalised). The joined 2,560-value vector is normalised and passed through two hidden layers (1,024 and 512 units, GELU, dropout 0.4) to two outputs. The figure also shows the cleaning and augmentation steps of the design; at prediction time the service applies

tokenisation, resizing and normalisation. The headline is stored for display, while the body text is the model's text input.

Table 5.1: Multimodal Model Configuration

Item	Value
Text encoder	csebuetnlp/banglabert, [CLS] vector (768)
Image encoder	EfficientNet-B4 (timm), average pooled (1,792)
Fusion and classifier	Concatenation (2,560) → LayerNorm → 1,024 → 512 → 2
Training	35 epochs, learning rate 2e-4, weight decay 0.01, batch size 6, label smoothing 0.1, 3 warm-up epochs
Output	FAKE or NON_FAKE with confidence values, shown as Likely Fake or Likely Real

5.1.2 Recorded Evaluation

The training notebook records the results in Table 5.2. The best validation accuracy during training was 82.92%.

Table 5.2: Recorded Test Results of the Multimodal Model (960 test items)

Class	Precision	Recall	F1-score	Support
Real (0)	0.79	0.81	0.80	480
Fake (1)	0.81	0.79	0.80	480
Weighted average	0.80	0.80	0.80	960
Accuracy			0.80	960

5.2 Source-Based Verification Pipeline

Text and verified sources based verification runs a 13-stage pipeline (S01–S13), shown in Figure 5.2. A single context object is passed through every stage.

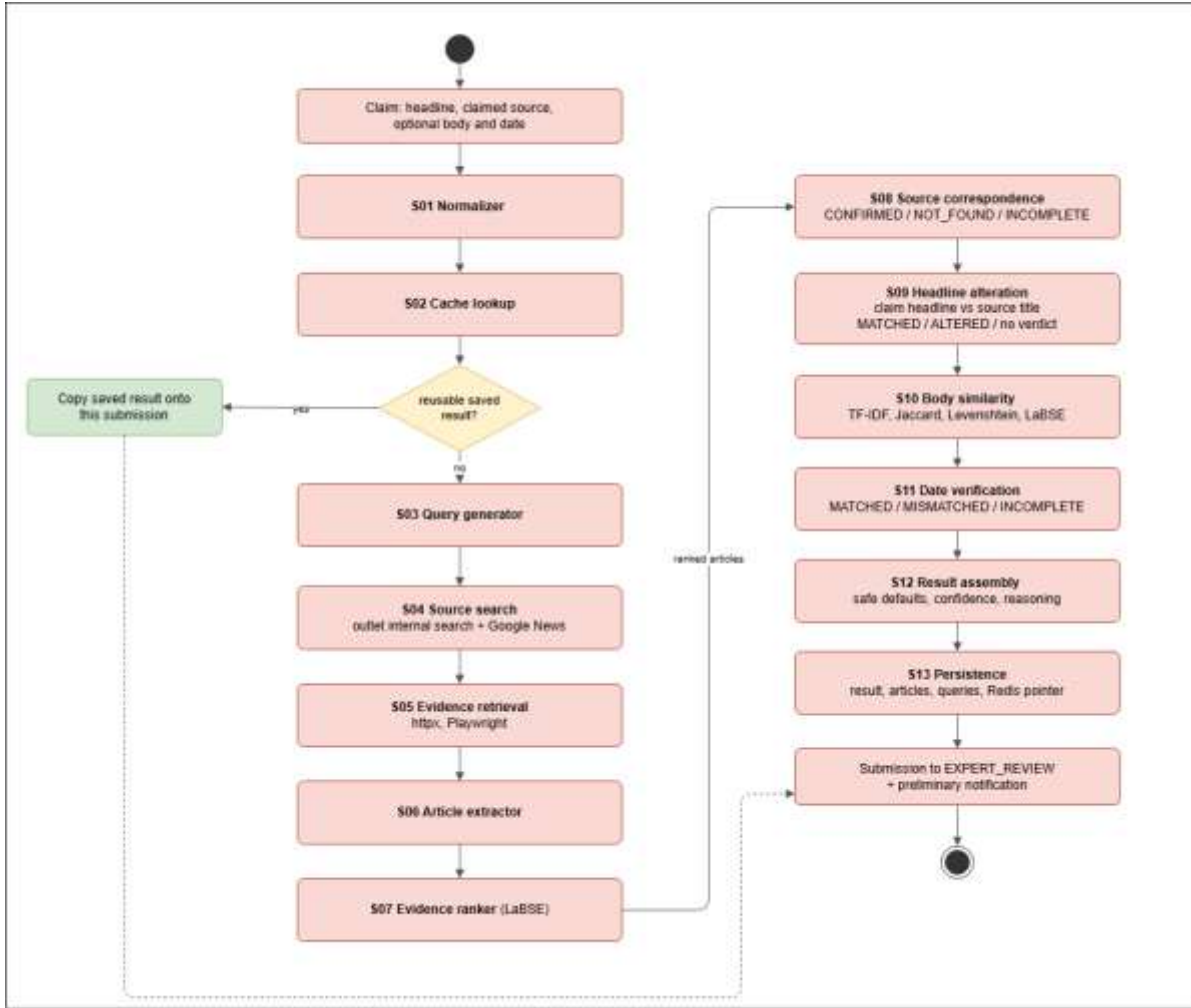


Figure 5.2: Source-Based Verification Pipeline — Stages S01 to S13

Table 5.3: Source-Based Verification Pipeline — Stage Reference

Stage	What it does
S01 Normalizer	Cleans the Bangla text, resolves the claimed outlet and builds the claim identity (hash).
S02 Cache lookup	Reuses a complete saved result for the same claim and pipeline version.
S03 Query generator	Builds up to six outlet-restricted queries from the headline and its keywords (dated variants when a date is given).
S04 Source search	Runs the outlet's own site search and Google News and rejects off-source results.
S05 Evidence retrieval	Fetches pages with httpx and uses Playwright for JavaScript pages.
S06 Article extractor	Extracts title, body and date with outlet selectors and general extractors.

Stage	What it does
S07 Evidence ranker	Ranks candidates with LaBSE similarity and keyword, date and domain signals; re-ranks with a cross-encoder when more than three remain.
S08 Source correspondence	Decides Confirmed, Not Found or Incomplete and selects the source article.
S09 Headline alteration	Compares the claim headline with the source title only: exact match, rule checks and NLI + LaBSE; Matched, Altered or no verdict.
S10 Body similarity	TF-IDF cosine, Jaccard, normalised Levenshtein and LaBSE cosine (only when a body is given).
S11 Date verification	Compares the claimed date with the article's date (Asia/Dhaka calendar).
S12 Result assembly	Builds the result, reasoning and analysis details; never an overall verdict.
S13 Persistence	Saves the result, articles and queries, sets the Redis pointer and notifies the submitter once.

Headline alteration uses deterministic rules for material changes, such as different numbers, dates, negation, attribution, subject–object roles and named entities. A rule-detected change gives Altered. A Matched result needs positive evidence that the meaning is the same; when neither can be shown, the result gives no verdict and states the reason. The body-similarity scores are measurements only and are never used as a verdict.

5.3 Model Selection

Table 5.4 lists the models used and why each was chosen. Cheap, cacheable models are used widely and the more expensive cross-encoders run only on a few selected pairs.

Table 5.4: AI Model Selection Summary

Model	Used in	Reason
BanglaBERT	Text & image (text encoder)	Pretrained on Bangla; understands Bangla vocabulary and grammar.
EfficientNet-B4	Text & image (image encoder)	Good accuracy for its computing cost.
LaBSE (sentence-transformers)	S07–S10	Multilingual sentence embeddings that cover Bangla; embeddings are cached by text hash.
mMARCO MiniLM cross-encoder	S07 re-ranking	More precise ranking of claim–article pairs; used only when more than three candidates remain.
mDeBERTa-v3 NLI (MNLI/XNLI)	S09	Detects entailment and contradiction between the headline and the source title in both directions.
BanglaTag NER	S09 entity checks	Bangla named-entity model used to spot changed persons, places and organisations.

5.4 Responsible Use of AI

- Automated output is always labelled preliminary; the overall verdict field stays empty until experts or an administrator finalize it.

- A source that was not found is not evidence of fake news and an incomplete search is never shown as a confident negative.
- Model limits are stated openly: the NLI model was not trained on Bangla, the NER model passed only a label and smoke check and the similarity thresholds are hand-set without a labelled Bangla benchmark. The pipeline tests use deterministic fakes and prove the logic, not model accuracy.

6. Component-Level Design

This chapter refines the analysis models into components, design classes, data stores, behaviour and deployment. Every name used here exists in the source code.

6.1 Component Diagram

Figure 6.1 shows the main components. Two clients, the Angular website and the extension, call one FastAPI backend. Inside the backend, feature modules handle requests, background workers run long work and shared services wrap the models, storage and cache. PostgreSQL, Redis and MinIO hold the data.

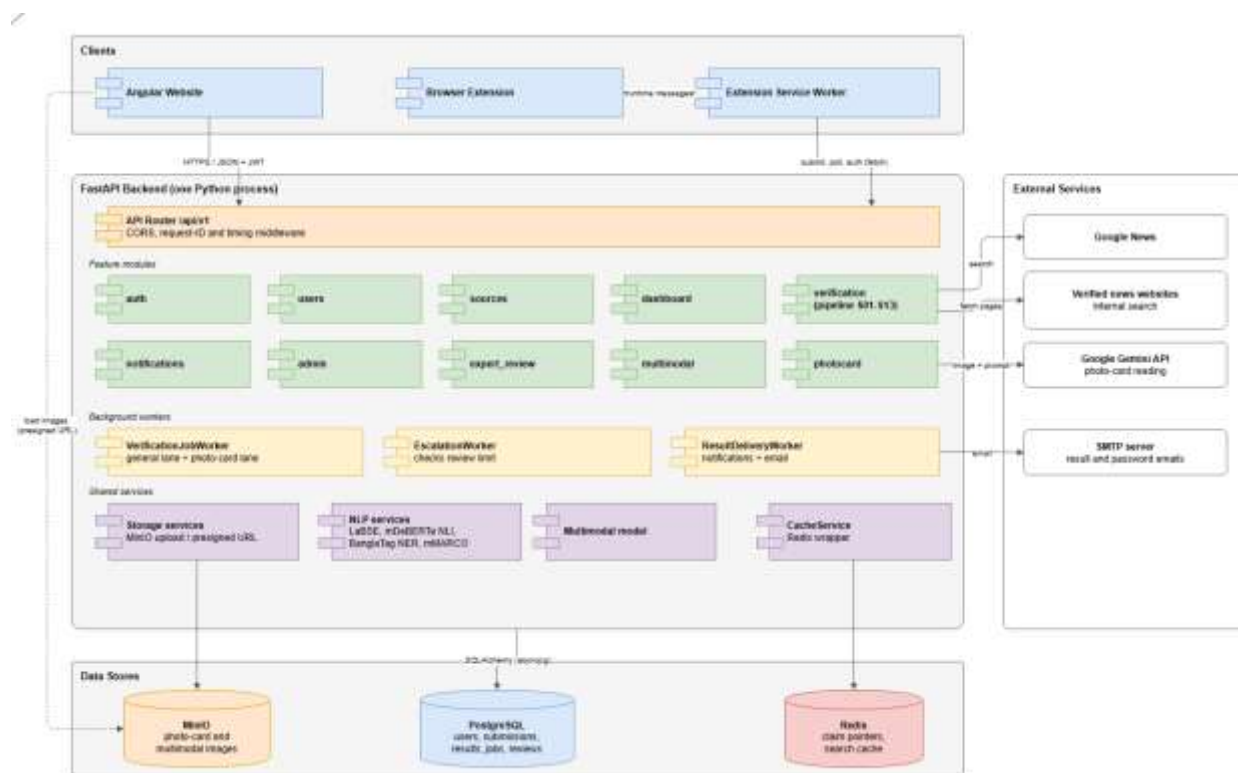


Figure 6.1: Component Diagram of BanglaFactGuard

Table 6.1: Main Components and Their Responsibilities

Component	Responsibility
Angular website	Public pages, claim forms, result reports, history, notifications, expert and admin workspaces.
Browser extension + service worker	Side panel for submission; the worker submits, polls, stores activity and raises alerts.
API router (/api/v1)	Routes requests; adds CORS, request ID and timing middleware.
auth, users	Registration, login, token refresh, OTP reset, profile and own submissions.
verification	Text & source claims and the S01–S13 pipeline.
photocard	Accepts card images, reads them with Gemini and runs the pipeline.
multimodal	Accepts text and image, detects duplicates and predicts.

Component	Responsibility
expert_review	Queue, votes, weighted tally, finalization and escalation.
admin, sources	Experts, verified sources, tiers, voting configuration and dashboard.
dashboard, notifications	Fact Explorer data and statistics; in-app notifications.
VerificationJobWorker	Drains the durable job queue in two lanes (general and photo card).
EscalationWorker	Every 60 s, escalates claims that reached a review limit.
ResultDeliveryWorker	Every 15 s, delivers result notifications and final-result emails.
Shared services	MinIO storage, NLP services, multimodal model and Redis CacheService.

6.1.1 Information Flow

A Text & source claim moves through the components as follows:

1. The website sends POST /verify/async with the claim and a JWT token (if signed in).
2. The verification module resolves the outlet and saves the Submission (PENDING) and its job in one transaction, then returns 202 Accepted with the submission ID.
3. The website polls GET /verify/{id}/status every 4 seconds while it stays open.
4. VerificationJobWorker claims the job with FOR UPDATE SKIP LOCKED and runs the pipeline.
5. The pipeline searches the outlet and Google News, fetches and ranks articles and decides each finding with the NLP services.
6. S13 saves the result, sets the status to EXPERT_REVIEW and creates a preliminary notification.
7. Experts vote through the expert_review module; when the rules pass, the verdict is saved and ResultDeliveryWorker notifies and emails the submitter.

6.2 Design Classes

Figure 6.2 shows the design classes that implement the problem domain and Table 6.2 maps each domain concept to its class.

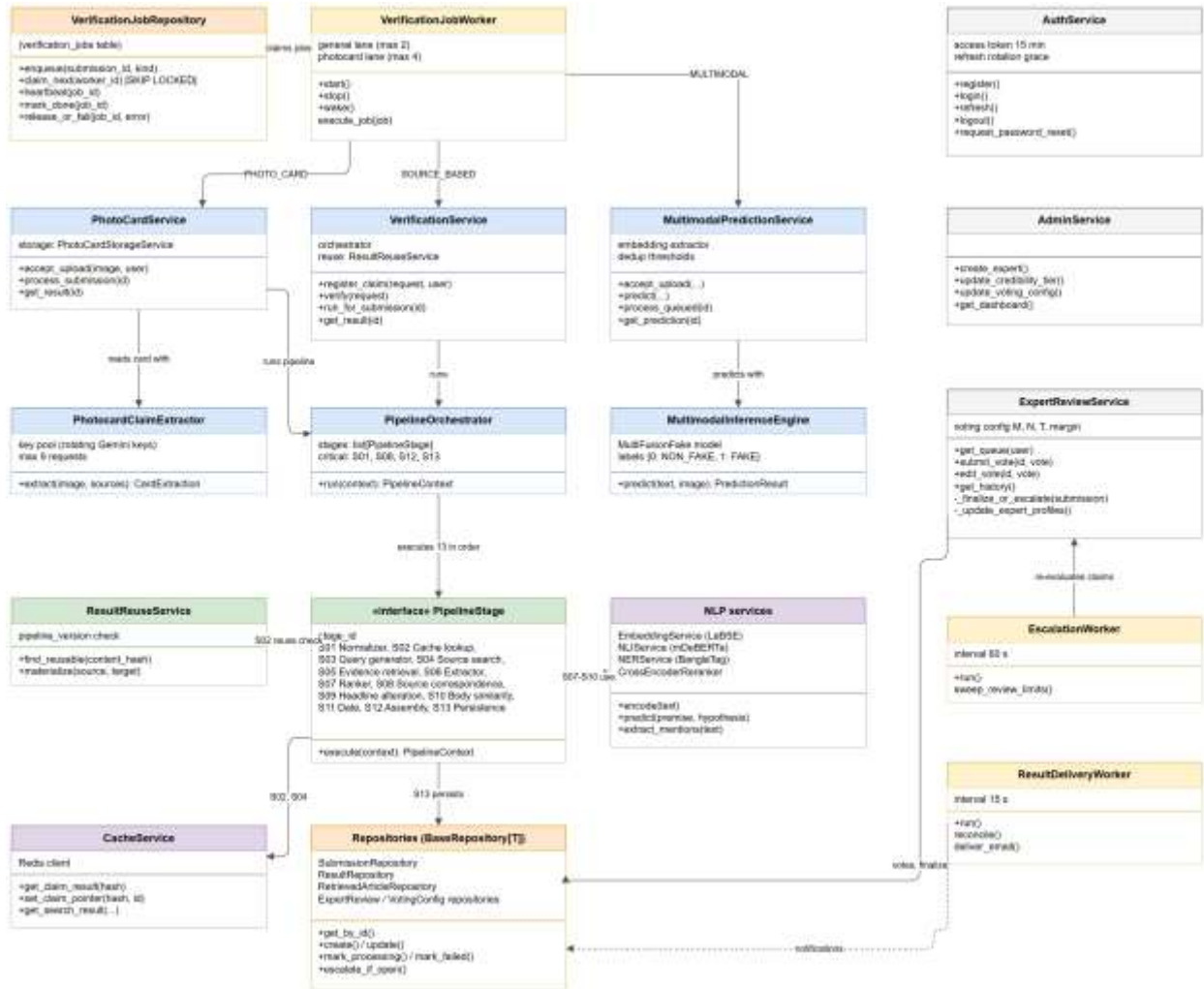


Figure 6.2: Design Class Diagram

Table 6.2: Design Classes and Domain Concepts

Domain concept	Design class	Main operations
Claim checking	VerificationService, PipelineOrchestrator, PipelineStage (S01–S13)	register_claim(), run_for_submission(), run(context), execute(context)
Photo card	PhotoCardService, PhotocardClaimExtractor	accept_upload(), process_submission(), extract(image, sources)
Text & image	MultimodalPredictionService, MultimodalInferenceEngine	accept_upload(), predict(text, image)
Background work	VerificationJobRepository, VerificationJobWorker	enqueue(), claim_next(), heartbeat(), mark_done()
Result reuse	ResultReuseService, CacheService	find_reusable(), materialize(), get_claim_result()
Expert voting	ExpertReviewService, EscalationWorker	submit_vote(), _finalize_or_escalate(), sweep_review_limits()
Accounts and admin	AuthService, AdminService	login(), refresh(), create_expert(), update_voting_config()
Notifications	notify_once(), ResultDeliveryWorker	reconcile(), deliver_email()
Data access	BaseRepository[T] and repositories	get_by_id(), create(), update(), mark_failed(), escalate_if_open()

6.3 Persistent Data Sources

Figure 6.3 shows which backend classes write to which data store and Table 6.3 describes each store.

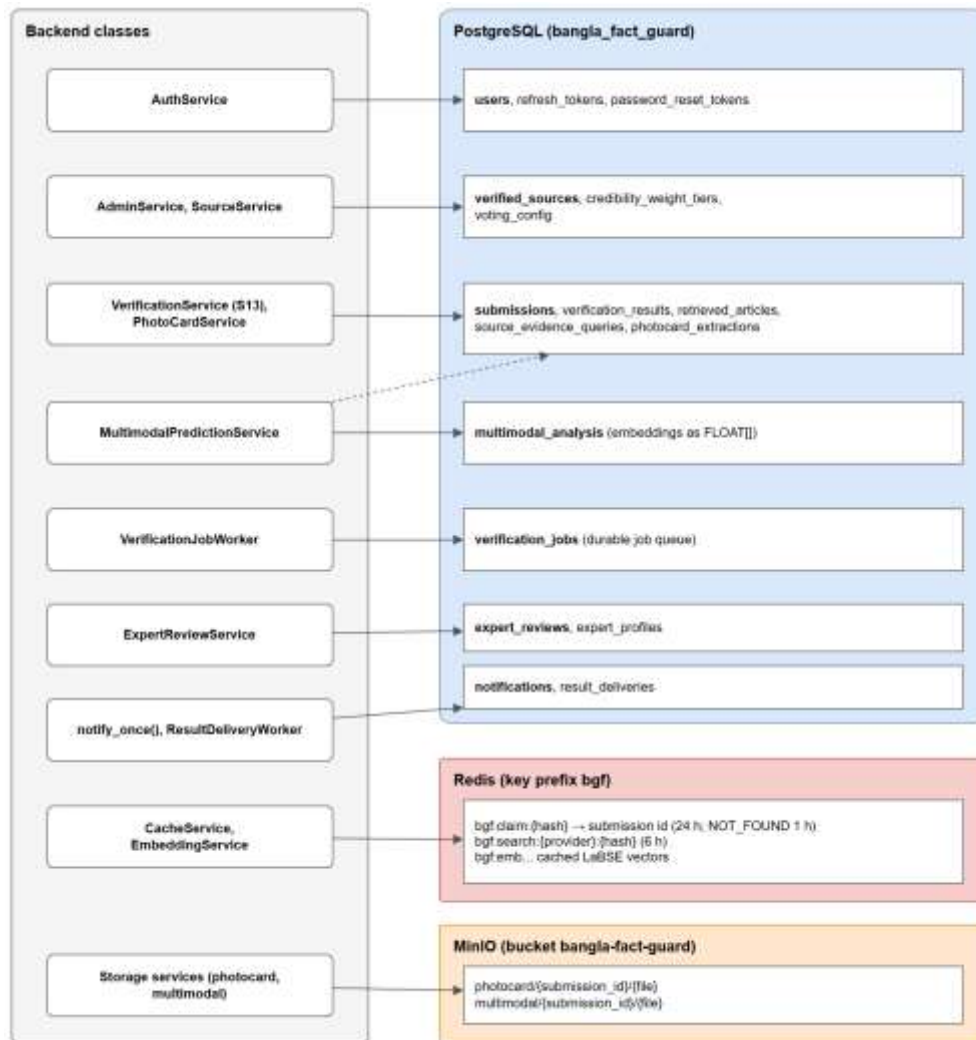


Figure 6.3: Persistent Data Sources and Their Classes

Table 6.3: Persistent Data Sources

Data store	What is stored	Lifetime
PostgreSQL (bangla_fact_guard)	Users, tokens, sources, submissions, results, articles, queries, extractions, analyses, reviews, notifications, the job queue (verification_jobs) and result deliveries	Permanent; migrations managed by Alembic
Redis (key prefix bgf)	Claim pointers to the submission holding a result, search results and LaBSE embeddings	Claim pointer 24 h (Not Found 1 h); search 6 h
MinIO (bucket bangla-fact-guard)	Photo card and multimodal images, served by presigned URLs	Permanent
Browser storage (extension)	Settings, tokens, drafts and account-separated activity	Until cleared or signed out

6.4 Behavioural Design

6.4.1 Submission Lifecycle

Every submission follows the state machine in Figure 6.4. The status is stored in the submissions table and the processing phase shows finer progress.

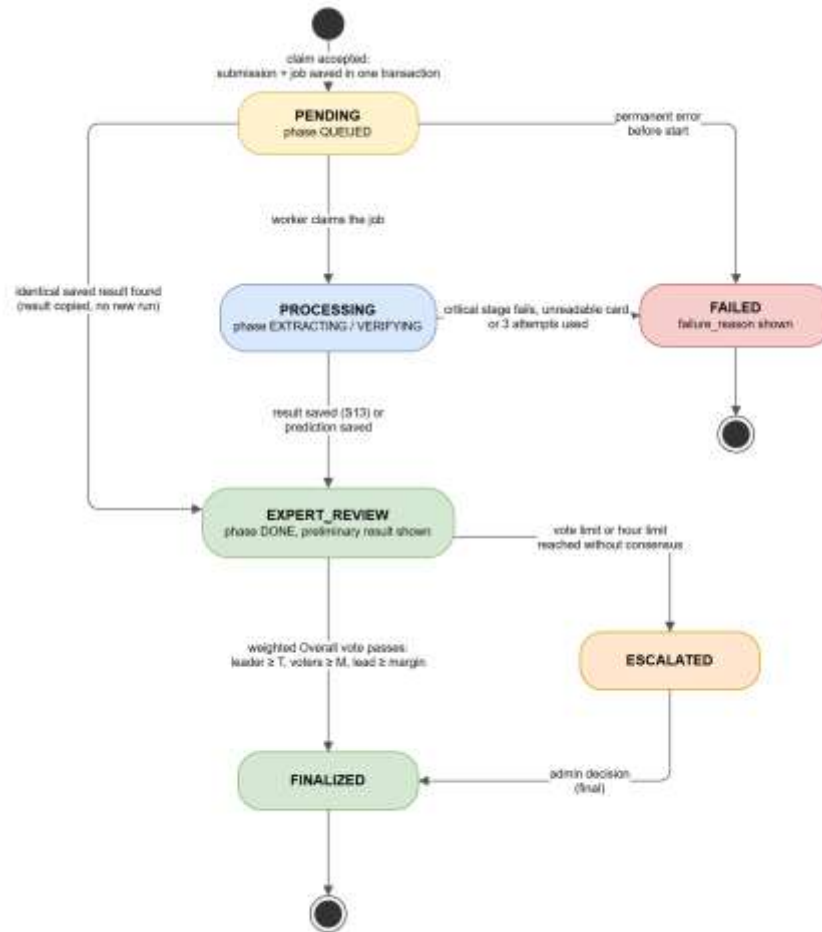


Figure 6.4: State Diagram of a Submission

Table 6.4: Submission States

State	Meaning	Leaves when
PENDING	Claim accepted; job queued (phase QUEUED)	A worker claims the job, a saved result is copied or a permanent error occurs
PROCESSING	Phase EXTRACTING or VERIFYING	The result or prediction is saved or a critical stage fails after up to 3 attempts
EXPERT_REVIEW	Preliminary result shown (phase DONE)	The weighted vote passes or a vote or time limit is reached
ESCALATED	Waiting for an administrator	The administrator submits the final decision
FINALIZED	Final verdict published	End state
FAILED	failure_reason shown to the user	End state

6.4.2 Sequence of a Text & Source Check

Figure 6.5 shows the full sequence from submission to expert vote, including the 4-second polling loop, the worker, the pipeline and the later final notification.

6.4.3 Voting Rules

Each vote is weighted. A new expert's vote has weight 1 until the expert has N completed reviews (the activation threshold); after that the weight comes from the accuracy tier set by the administrator (for example Novice 0.5, Competent 1, Expert 1.5 and Master 2 in the demonstration setup). A claim is finalized only when all of the following hold: there is one clear leading verdict, its weight is at least T , at least M experts voted and it leads the runner-up by the margin. A tie never finalizes. If the maximum number of votes or hours is reached first, the claim is escalated and the administrator's decision is final.

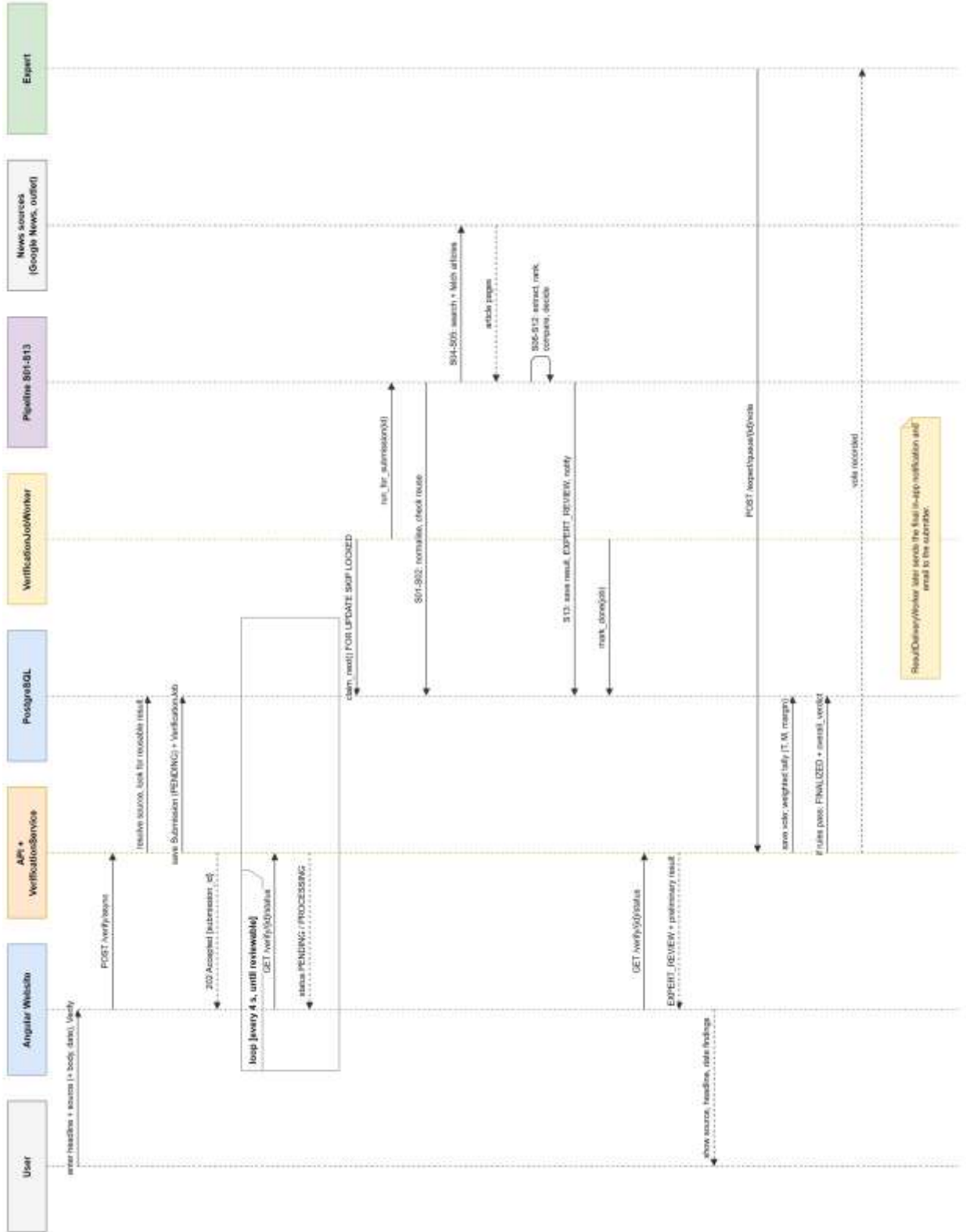


Figure 6.5: Sequence Diagram — Text & Source Claim (rotated to fit the page)

6.5 Deployment Design

Figure 6.6 shows the deployment used for development and the demonstration. The browser loads the website from the Angular development server and calls the FastAPI server. PostgreSQL 17 runs on the host, while Redis and MinIO run in Docker Compose. Google News, Verified news websites, Gemini API, and Gmail SMTP are external services.

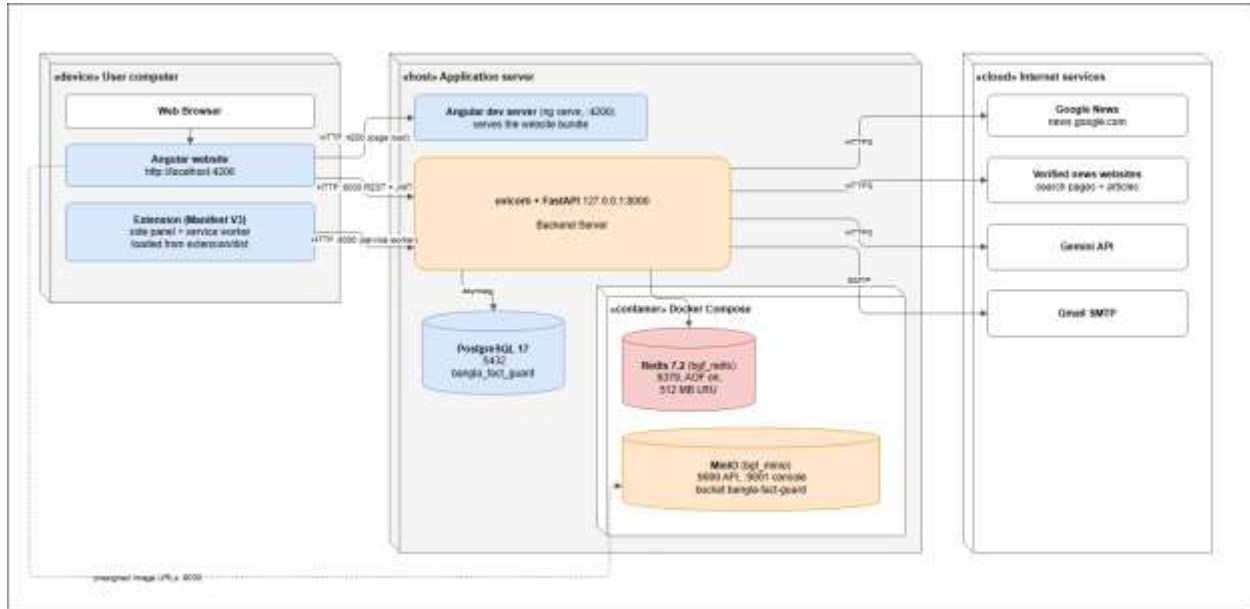


Figure 6.6: Deployment Diagram

Table 6.5: Deployment Nodes

Node	Contents	Notes
User computer	Chrome with the website and the unpacked extension (extension/dist)	Extension calls the API on port 8000
Application server	Angular dev server (:4200); uvicorn + FastAPI (127.0.0.1:8000) with the job, escalation and delivery workers	Workers run inside the API process
PostgreSQL 17	Database bangla_fact_guard (:5432)	Installed on the host, not in Compose
Docker Compose	Redis 7.2 (:6379, AOF, 512 MB LRU) and MinIO (:9000 API, :9001 console)	Started with docker compose up -d
Internet services	Google News, verified news websites, Gemini API, Gmail SMTP	Reached over HTTPS / SMTP

6.6 Design Alternatives Considered

Each component design was reviewed and refactored during development. The main alternatives and the reasons for the final choices are listed below.

Table 6.6: Design Alternatives and Decisions

Decision	Alternative considered	Reason for the choice
Durable job table in PostgreSQL	Separate broker (Celery with Redis)	Jobs are saved in the same transaction as the submission and survive restarts without another service.
Separate Source, Headline and Date findings	One automated True/False/Partially True label	A single label hid why a claim failed and mixed missing evidence with falsehood.
Headline compared with the source title only	Headline compared with the whole article body	Body text often repeats or rephrases facts and gave misleading matches.
Gemini reading of photocards	Tesseract OCR with manual correction	OCR could not reliably read stylised Bangla cards or identify the outlet logo.
Redis stores only a pointer; the database is authoritative	Cache full results in Redis	Prevents stale or incompatible cached scores from overriding saved results.
In-process workers	Separate worker processes	Simpler deployment for the project; the queue still allows more workers later.

7. Interface Design

BanglaFactGuard has two user interfaces: the website, used by all four roles and the extension side panel, used to submit and track claims while browsing. This chapter defines the interface objects and actions, the navigation, the events that change the interface state and how each state looks to the user.

7.1 User Interface Objects and Actions

Table 7.1: User Interface Objects and Actions

Interface	UI object	Actions
Header	Navigation links, bell icon, user menu	Open Verify facts, Fact Explorer, My Submissions, Expert Queue, Administration; open notifications; sign out
Verify facts	Method cards	Choose News story against an image, News story against claimed source or Photocard against claimed source
Verify facts	Claim form	Type or paste headline and text; choose outlet and date; upload image; Verify
Verification result	Report sections	Read claim, final decision, preliminary result, article, headline comparison, body scores and date; open the original source; copy the report link
Fact Explorer	Statistics, tabs, search and filters	Switch Final decisions / Under review; search; filter by decision, type, date and outlet; page through results
My Submissions, Notifications	Lists	View details; mark as read; mark all as read
Expert workspace	Queue, review page, vote form	Search; Review & vote; choose verdict; add optional supplementary findings; write justification; Submit vote
Administration	Dashboard, tables and forms	Create, edit, deactivate experts and reset passwords; register, edit, disable and delete sources; edit tiers and voting rules; decide escalated claims
Extension	ON/OFF switch, Verify, Activity and Account tabs	Use selected text; upload image; select screenshot area; crop; submit; refresh; mark read; view details; sign in/out

7.2 Navigation

Figure 7.1 shows how the website screens are connected for each role. Every role reaches the same verification result page, which is the shared report for all three methods.

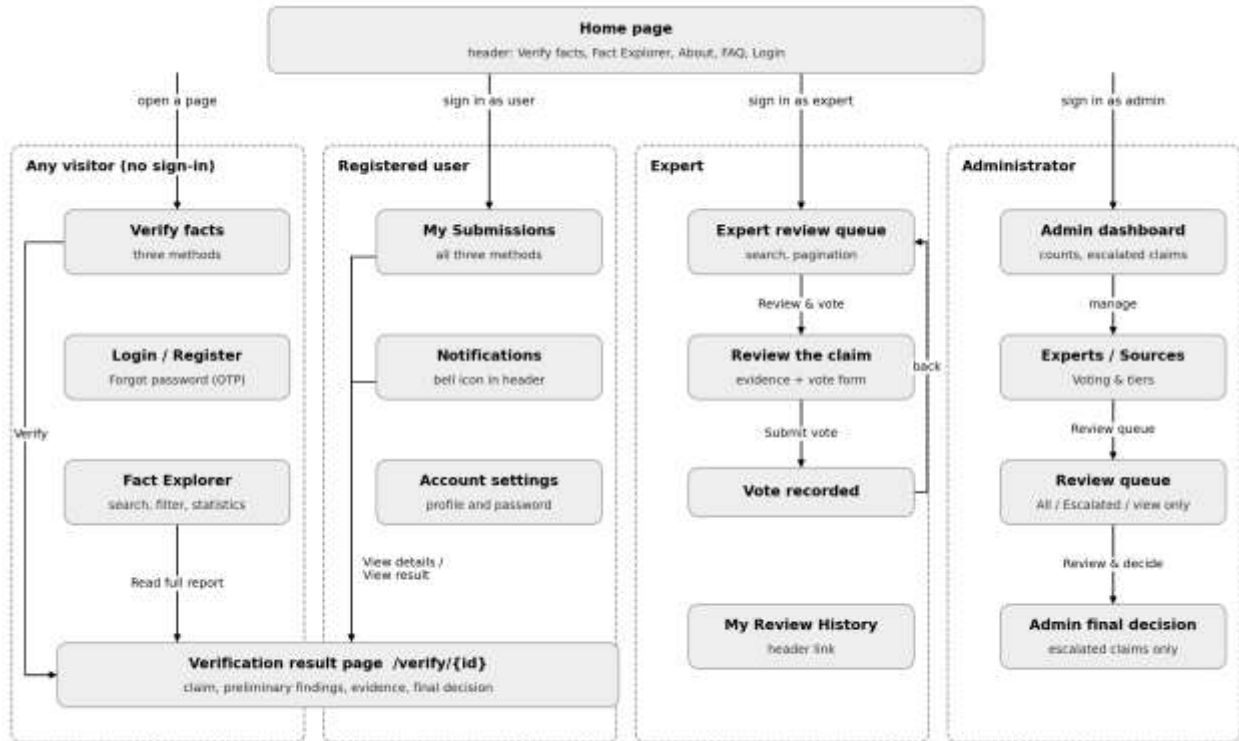


Figure 7.1: Website Navigation by User Role

7.3 UI Events and State Transitions

7.3.1 Verify Facts Page

Figure 7.2 shows the states of the Verify facts page and Table 7.2 lists the user events that cause each change.

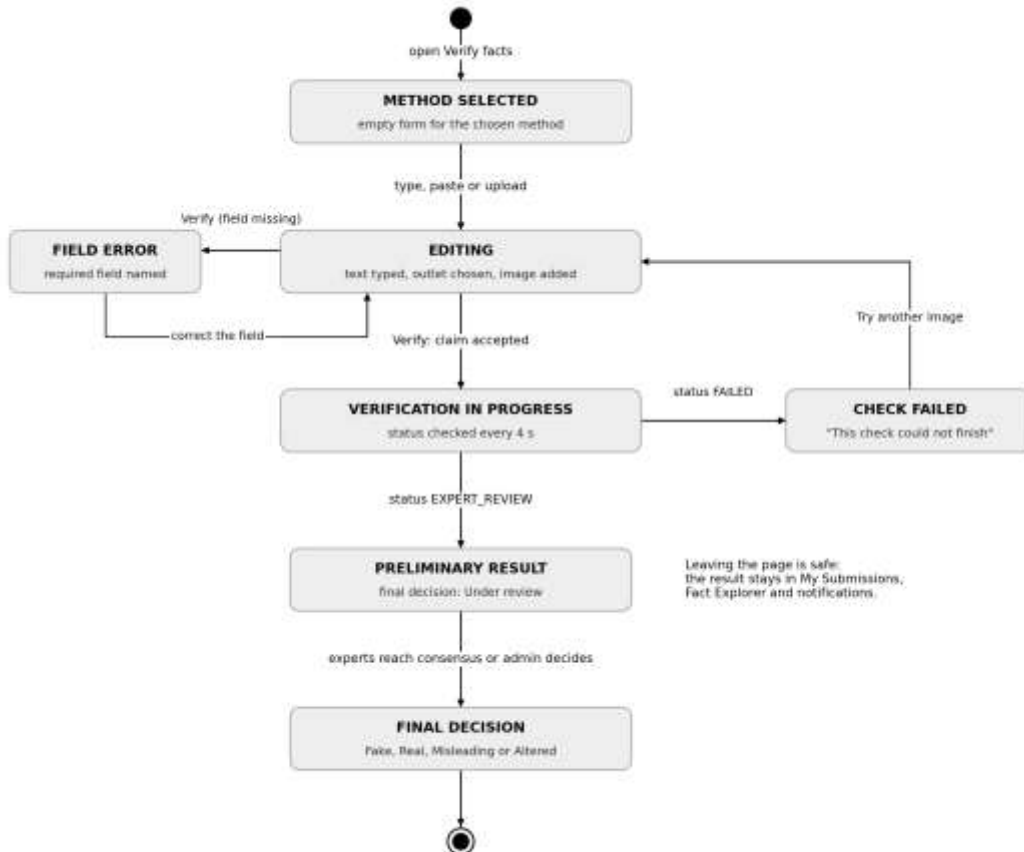


Figure 7.2: State Diagram of the Verify Facts Interface

Table 7.2: Events and State Transitions on the Verify Facts Page

User or system event	From state	To state	What the user sees
Choose a method card	none	METHOD SELECTED	The matching form and a preview panel
Type, paste or upload	METHOD SELECTED	EDITING	Live preview of the claim or image
Click Verify with a required field empty	EDITING	FIELD ERROR	The field is named and the inputs are kept
Click Verify with valid input	EDITING	IN PROGRESS	“Verification in progress” with View progress and My Submissions
Status poll returns EXPERT_REVIEW	IN PROGRESS	PRELIMINARY RESULT	Findings with Final decision: Under review
Status poll returns FAILED	IN PROGRESS	CHECK FAILED	“This check could not finish” with the reason
Experts or admin finalize	PRELIMINARY RESULT	FINAL DECISION	Final decision badge and Voting Details

7.3.2 Expert and Administrator Review

Figure 7.3 shows the review interface. The vote form keeps the user on the same state until the justification has at least 50 characters. An escalated claim appears on the administrator dashboard and only the administrator can decide it.

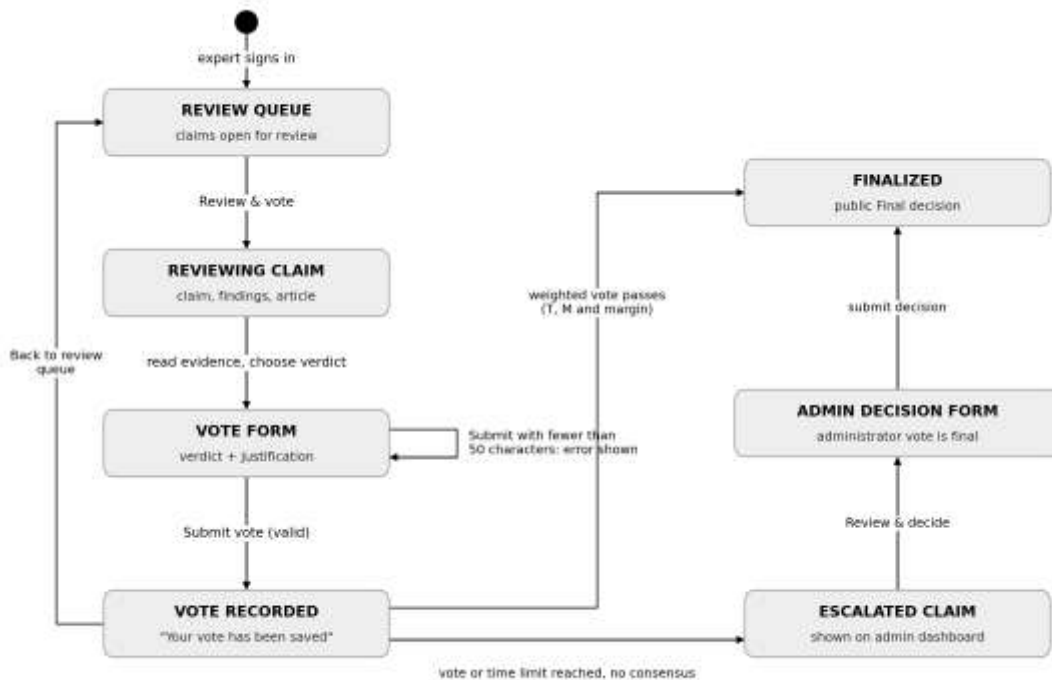


Figure 7.3: State Diagram of the Review Interface

7.3.3 Browser Extension Side Panel

Figure 7.4 shows the side panel states. Switching the extension OFF stops new submissions and alerts, but jobs already accepted by the server continue. The Account tab changes between guest and signed-in states independently of the Verify flow.

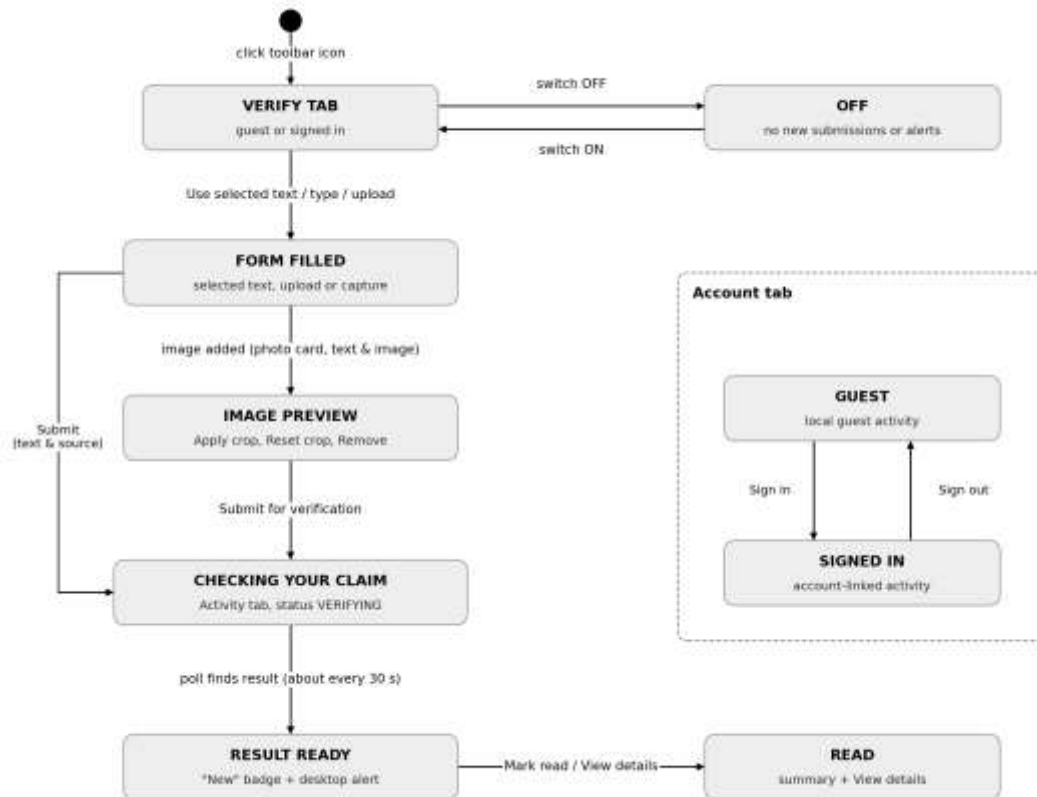


Figure 7.4: State Diagram of the Extension Side Panel

7.4 Interface States as Seen by the User

The figures below show the main interface states as they appear in the running system. Each screenshot is cropped to the part of the page that shows the state.

Figure 7.5 shows the EDITING state of a Text & source claim: the selected method card is highlighted, the form holds the headline, outlet and date and the preview on the right updates as

the user types. After **Verify news** is clicked, the page changes to the IN PROGRESS state in Figure 7.6.

BanglaFactGuard Home Verify facts Fact Explorer My Submissions About Demo User

Verify facts

Hide descriptions

☐ News story against an image

Check if the news story matches the given image. First an initial result will be provided by model prediction. Then experts will review it and give the final result.

☒ News story against claimed source

Check if the news story matches the original news from the claimed source. First an initial result will be provided by automatic evaluation. Then experts will review it and give the final result.

☐ Photocard against claimed source

Check if the photocard matches the original news from the claimed source. First an initial result will be provided by automatic evaluation. Then experts will review it and give the final result.

Provide News Details

Headline *

গাজীপুরে পোশাক কারখানায় আগুন, ফায়ার সার্ভিসের নয়টি ইউনিটের তিন ঘণ্টার চেষ্টায় পরিস্থিতি নিয়ন্ত্রণে

Claimed news outlet *

প্রথম আলো

Choose the outlet the claim says published this news.

Article text optional

Enter the news text.

Claimed publication date

10/03/2026

✓ Verify news

গাজীপুরে পোশাক কারখানায় আগুন, ফায়ার সার্ভিসের নয়টি ইউনিটের তিন ঘণ্টার চেষ্টায় পরিস্থিতি নিয়ন্ত্রণে

Claimed outlet: prothomalo.com Comparison scope: Headline only

Figure 7.5: EDITING State — Text & Source Claim Form

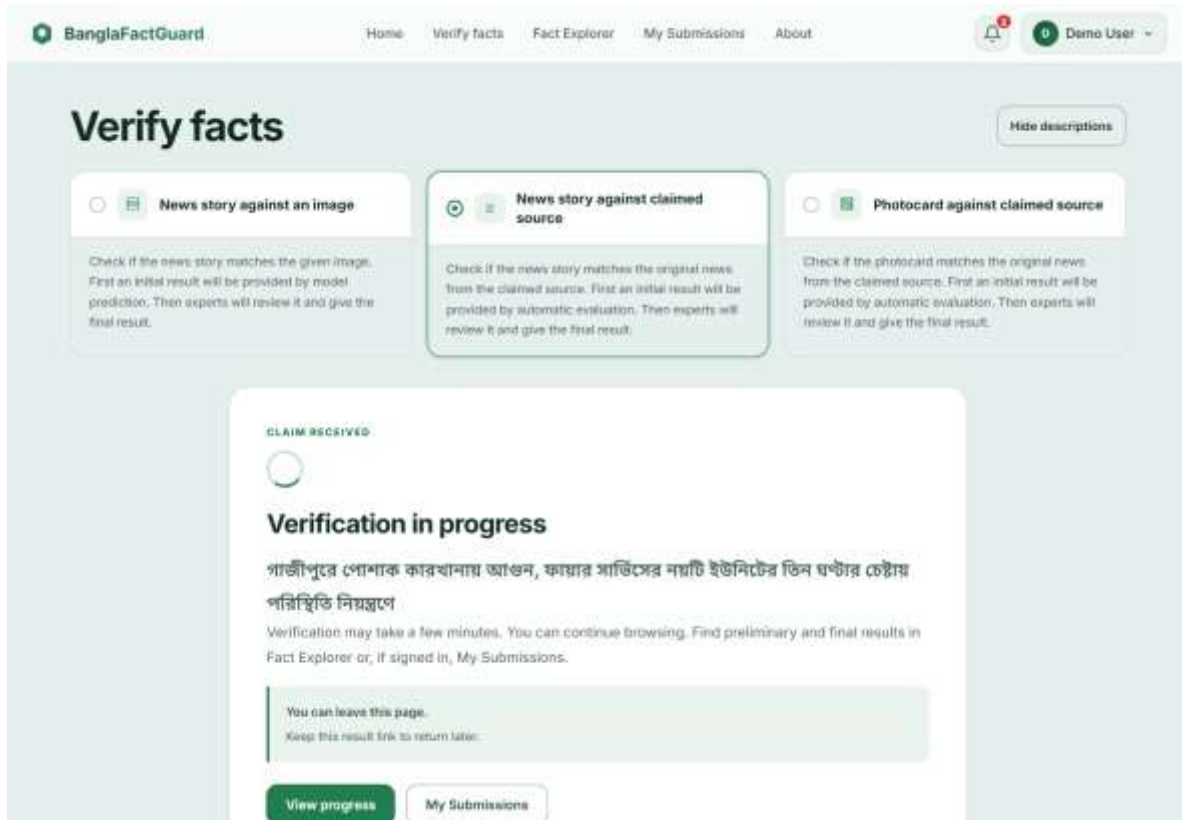


Figure 7.6: IN PROGRESS State — Claim Received

When the automated check is finished, the report page shows the PRELIMINARY RESULT state (Figure 7.7). The dashed Final decision box says Under review. After expert review, the same box becomes the FINAL DECISION state with a verdict badge and Voting Details (Figure 7.8).

The screenshot shows the BanglaFactGuard interface. At the top, there's a navigation bar with 'Home', 'Verify facts', and 'Fact Explorer'. A 'Login' button and a 'Get Started' button are on the right. Below the navigation bar, there's a 'Fact Explorer' button on the left and 'Start another check' and 'Copy report link' buttons on the right. The main heading is 'Verification result'.

Under 'Verification result', there are three main sections:

- SUBMITTED CLAIM:** গাজীপুরে পোশাক কারখানায় আগুন, ফায়ার সার্ভিসের নয়টি ইউনিটের তিন ঘণ্টার চেষ্টায় পরিস্থিতি নিয়ন্ত্রণে
- FINAL DECISION:** Under review. Expert review pending. No final decision has been made yet; the findings below are preliminary and automated.
- Preliminary result:** This section shows a comparison of the claim with a relevant article from a claimed source. It includes a table with three columns: '01 / RELEVANT ARTICLE FROM CLAIMED SOURCE', '02 / HEADLINE ALTERATION', and '03 / DATE'. The results are 'Found', 'Exact Matched', and 'Matched' respectively. Below the table, there are three explanatory notes for each result.

01 / RELEVANT ARTICLE FROM CLAIMED SOURCE	02 / HEADLINE ALTERATION	03 / DATE
Relevant article from claimed source: Found	Headline: Exact Matched	Date: Matched
A relevant article was found in the news outlet the claim names. This alone does not make the claim true. Claimed outlet: prothomalo.com	The claim headline is word-for-word the same as the source title (ignoring spacing and a final full stop).	The claimed publication date agrees with the source article's date.

Figure 7.7: PRELIMINARY RESULT State — Final Decision Under Review

BanglaFactGuard
Home
Verify facts
Fact Explorer
Login
Get Started

Fact Explorer
Start another check
Copy report link

Verification result

SUBMITTED CLAIM

আমিন খানের সঙ্গে যে কেউ কাজ করতে চায়...

FINAL DECISION

Final decision Altered

Expert reviewers could not agree, so an administrator made this final decision. The preliminary AI findings below are kept for reference; they are not the final decision.

Voting Details

Preliminary result

Comparison scope: Headline only

S1 / RELEVANT ARTICLE FROM CLAIMED SOURCE	S2 / HEADLINE ACTERATION	S3 / DATE
Relevant article from claimed source: Found A relevant article was found in the news outlet the claim names. This alone does not make the claim true. Claimed outlet: prothomalo.com	Headline: Altered The claim headline differs meaningfully from the source title. See the evidence below:	Date: Mismatched The claimed publication date differs from the source article's date. This alone does not mean the news is false.

Figure 7.8: FINAL DECISION State — Administrator Decision on a Photo Card

If a check cannot finish, the CHECK FAILED state explains the reason and offers a recovery action instead of a verdict (Figure 7.9). The extension shows the same life cycle in its Activity tab (Figure 7.10).

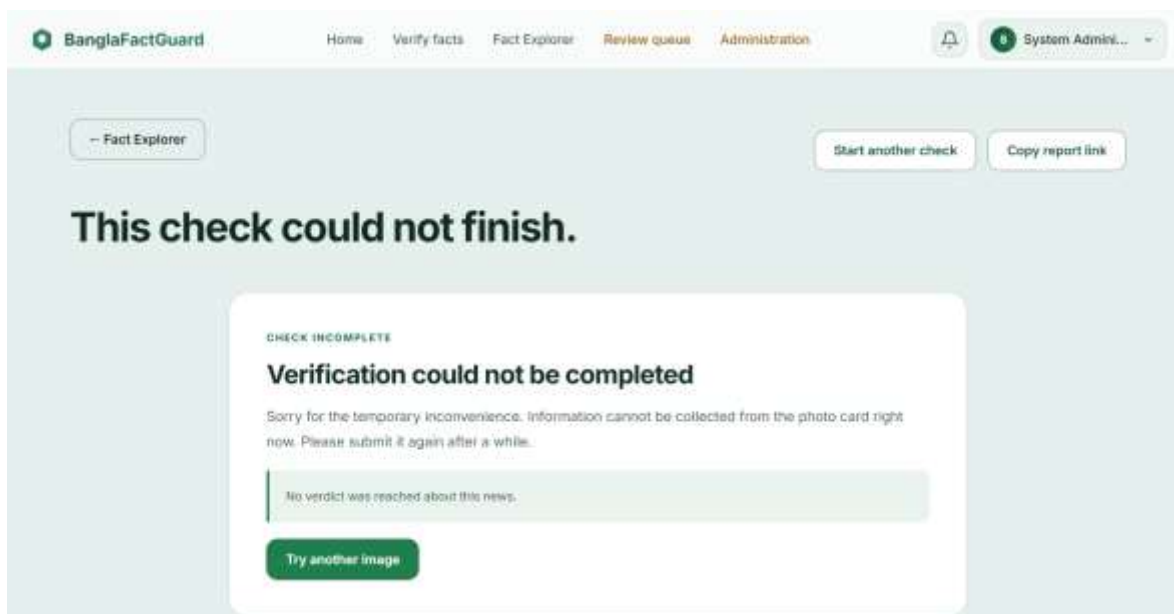


Figure 7.9: CHECK FAILED State — Photo Card Could Not Be Read

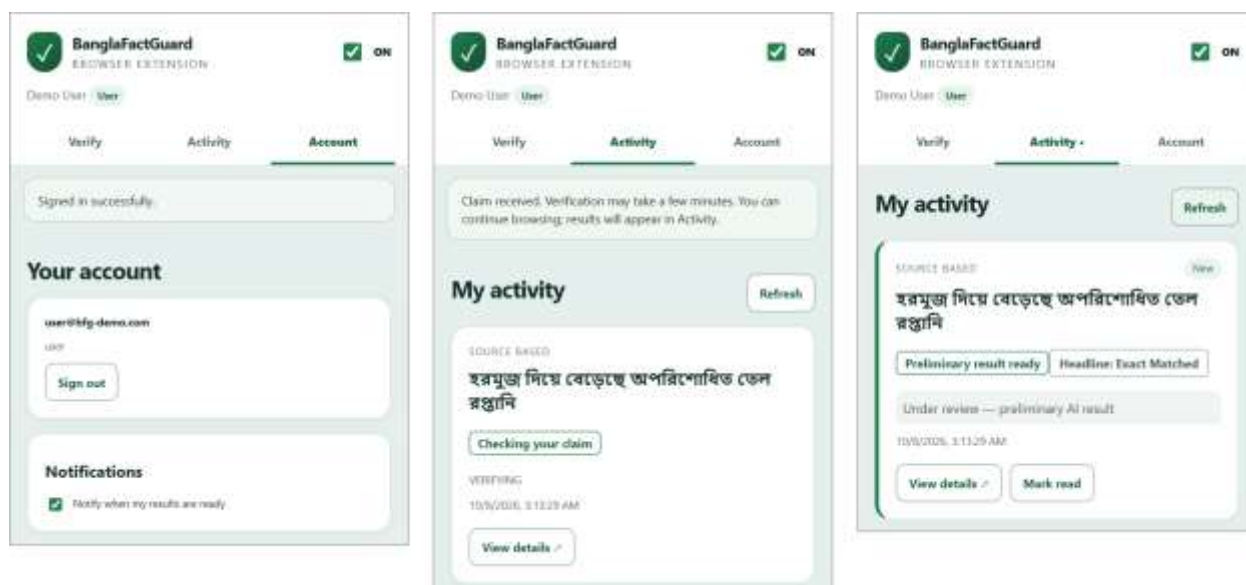


Figure 7.10: Extension States — Signed In, Checking Your Claim and Result Ready

Figure 7.11 shows the VOTE FORM state of the expert review page. The left column repeats the claim, the preliminary findings and the article; the right column holds the decision guidance and the vote form with the justification counter.

BanglaFactGuard Home Verify facts Fact Explorer **Expert Queue** Review History

[Review queue](#) [View public report](#)

EXPERT WORKSPACE

Review the claim. Follow the evidence.

Read the original submission and source material before making your independent assessment. Automated findings are supporting information, not instructions for your decision.

01 / ORIGINAL SUBMISSION

Read the claim in context

শিক্ষার্থী বিক্ষোভে উত্তাল ফ্রান্স, ২০০০ শিক্ষাপ্রতিষ্ঠান বন্ধ

Claimed outlet: prothomalo.com
Claimed publication date: Not provided
Submitted: Oct 6, 2025, 10:52:38 PM

YOUR INDEPENDENT REVIEW

Before you decide

1. Read the full claim and available source.
2. Distinguish missing evidence from a false statement.
3. Explain how the evidence supports your decision.

0 expert votes recorded. Only the overall vote decides the claim. After the final decision, each vote and its justification are shown publicly with the reviewer's name.

Preliminary result

Comparison scope: **Headline only**

01 / RELEVANT ARTICLE FROM CLAIMED SOURCE

Relevant article from claimed source: **Found**

A relevant article was found in the news outlet the claim names. This alone does not make the claim true.

Claimed outlet: prothomalo.com

02 / HEADLINE ALTERATION

Headline: **Exact Matched**

The claim headline is word-for-word the same as the source title (ignoring spacing and a final full stop).

Read the relevant article

Open the original publication and compare its wording with the submitted claim.

prothomalo.com

শিক্ষার্থী বিক্ষোভে উত্তাল ফ্রান্স, ২০০০ শিক্ষাপ্রতিষ্ঠান বন্ধ

Published 2025-10-08

03 / YOUR VOTE

Record your decision

Fields marked * are required.

Cast your vote based on your findings *

☒ Real ☐ Fake

☐ Misleading ☐ Altered

[Supplementary findings \(optional\)](#)

Justification * at least 50 characters, shown publicly after the final decision

I compared the claim headline with the source article title and the full report. The details shown support this verdict.

120 / 50-character's minimum

Submit vote

Figure 7.11: VOTE FORM State — Expert Review Page

8. Implementation

8.1 Technology Stack

Table 8.1: Technology Stack

Part	Technology
Website	Angular 19, TypeScript, SCSS, Angular Material/CDK
Backend	Python (3.11 or newer; 3.13.5 in the test environment), FastAPI, SQLAlchemy 2 (async, asyncpg), Alembic, Pydantic settings, structlog
AI and NLP	PyTorch, timm, Hugging Face Transformers, sentence-transformers, Google Gemini API (photo cards)
Retrieval	httpx, Playwright (headless Chromium), trafilatura, readability, BeautifulSoup, pygooglenews
Data	PostgreSQL 17, Redis 7.2, MinIO
Extension	Preact, Vite 6, Chrome Manifest V3 (side panel and service worker)
Testing	pytest 9 with SQLite helpers and fakes; Karma/Jasmine with headless Chrome; node:test with happy-dom and fake-indexeddb

8.2 Code Organisation

The repository has three applications. The backend (backend/app) is organised by feature: auth, users, submissions, verification (with pipeline/stages s01–s13), photocard, multimodal, expert_review, admin, sources, dashboard, notifications, search, nlp and cache. The website (frontend/src/app) has feature folders for home, verification, dashboard (Fact Explorer), history, notifications, settings, expert, admin, auth, about and faq. The extension (extension/src) contains the panel, the background worker, capture and shared helpers.

8.3 Key Implementation Details

8.3.1 Authentication and Roles

Passwords are hashed and access tokens are short-lived (15 minutes). The website refreshes the token before it expires and shares one refresh between parallel requests; a temporary network failure does not sign the user out. Password reset uses a 6-digit code that expires after 10 minutes. Role checks protect the expert and admin routes.

8.3.2 Background Jobs and Reuse

Every accepted claim is saved with a row in verification_jobs. The in-process worker claims jobs with FOR UPDATE SKIP LOCKED, sends a heartbeat every 30 seconds and re-claims jobs whose heartbeat is older than 120 seconds. Photo card jobs have their own lane (up to 4 at once) so that waiting for Gemini never blocks text jobs (up to 2 at once). A failed job is retried up to three times; permanent errors fail at once with a readable reason. A claim that was already checked gets a copy of the saved result on its own submission and the copy follows the original's final verdict.

8.3.3 Photo Card Reading

Gemini reads the headline exactly as printed, chooses the outlet only from the active verified sources and reads the printed date without guessing. At most nine requests are made (three batches of three, 10 seconds apart). If none succeeds or no headline or known outlet is found, the submission fails with a message that tells the user what to do.

8.3.4 Voting, Escalation and Notifications

ExpertReviewService locks the submission row before saving a vote, rejects a second vote by the same expert and blocks experts from reviewing their own claims. The weighted tally ignores the AI output. EscalationWorker checks the review limits every minute. notify_once() avoids duplicate notifications and ResultDeliveryWorker sends final-result emails with retry and backoff when SMTP is configured.

8.3.5 Browser Extension

The side panel stores drafts and activity per account. The service worker submits through the asynchronous endpoints, checks pending results about every 30 seconds (with backoff after network errors), shows desktop alerts and an unread badge and forgets a claim only when the server says that this submission no longer exists. Screenshot capture takes the visible tab and immediately crops it to the selected area. Only an administrator can change the API and website addresses.

8.4 Main API Routes

Table 8.2: Main API Routes (prefix /api/v1)

Operation	Route
Register, login, refresh, logout, profile	POST /auth/register, /auth/login, /auth/refresh, /auth/logout; GET /auth/me
Active verified sources	GET /sources
Queue a text claim; status and result	POST /verify/async; GET /verify/{id}/status, /verify/{id}
Queue a photo card; status and result	POST /photocard/verify/async; GET /photocard/{id}
Queue a text & image claim; result	POST /multimodal/predict/async; GET /multimodal/by-submission/{id}
Submission lookup; own submissions	GET /submissions/{id}; GET /users/me/submissions
Fact Explorer	GET /dashboard/explorer and statistics routes
Expert queue and vote	GET /expert/queue; POST /expert/queue/{id}/vote
Notifications	GET /notifications

9. Testing

9.1 Testing Approach and Strategy

Testing was done at three levels. **Unit and component tests** check each backend stage, service and rule with deterministic fakes for the models, search and storage. **Integration-style tests** run real database code on SQLite (only the needed tables are created) and run whole pipelines, the job worker and the voting service end to end with fakes for external services. **Client tests** check the website components and the extension panel and service worker with simulated browser APIs. Finally, the full system was used manually with real services to capture the screenshots in this report.

The strategy covers valid input, invalid input and failure situations for every main workflow: missing fields, wrong file types, an unknown outlet, failed searches, unreadable photo cards, ties and limits in voting, expired tokens and network failures.

9.2 Test Environment and Tools

Table 9.1: Test Environment

Item	Value
Execution date	8 October 2026
Machine	Windows 11 Home (10.0.26100)
Backend	Python 3.13.5, pytest 9.1.1; command: <code>python -m pytest tests/unit -q --no-cov</code>
Website	Node 22.17, Angular 19, Karma with Chrome Headless 154; command: <code>ng test --watch=false --browsers=ChromeHeadless</code>
Extension	Node 22.17, node:test with happy-dom and fake-indexeddb; command: <code>npm test</code>

9.3 Item Pass/Fail Criteria

- **Pass:** the test runs and every assertion holds (expected status, fields, saved state and messages).
- **Fail:** any assertion does not hold or the code raises an unexpected error.
- **Skipped / not executed:** the test needs a resource that was not enabled in the run (for example real model weights). It counts as neither pass nor fail.
- A release is acceptable when all executed automated tests pass. Model accuracy is reported separately and is not a pass/fail criterion of the software tests.

9.4 Summary of Executed Test Runs

Table 9.2 gives the results of the three suites run for this report.

Table 9.2: Executed Automated Test Runs (8 October 2026)

Suite	Tests	Passed	Failed	Skipped	Time
Backend unit tests (44 files)	426	424	0	2	20.4 s
Website (16 spec files)	93	93	0	0	1.8 s
Extension (3 test files)	33	33	0	0	2.7 s
Total	552	550	0	2	

The two skipped backend tests run only with real local model weights (BFG_REAL_MODEL_TESTS=1); they were not enabled in this run. Error lines that appear in the backend log come from tests that deliberately simulate failures and are expected.

9.5 Test Cases with Detailed Outcomes

The tables below list the important test cases. Cases without a mark are **automated**: they were run on 8 October 2026 and the actual result is the outcome of that run. Cases marked **(M)** are **manual**: they were carried out in the running system and their evidence is the screenshot named in the table.

9.5.1 Authentication and Sessions

Table 9.3: Test Cases — Authentication and Sessions

ID	Scenario	Input / steps	Expected result	Actual result	Status
TC-01	Refresh access token	Expired access token, valid refresh token	New token issued; user stays signed in	As expected	Pass
TC-02	Lost refresh response	Refresh succeeds but the reply is lost	User is not logged out within the grace window	As expected	Pass
TC-03	Logout	Sign out, then use the old refresh token	Token revoked immediately	As expected	Pass
TC-04	Parallel 401s on website	Several requests fail with 401 together	One shared refresh, then all requests retried	As expected	Pass
TC-05	Temporary refresh failure	Network error during refresh	Session is kept; sign-in required only when refresh is rejected	As expected	Pass
TC-06 (M)	Register form validation	Password shorter than 8 characters	Inline message “At least 8 characters”	Message shown (Figure 10.2)	Pass

9.5.2 Text & Source Verification

Table 9.4: Test Cases — Text & Source Verification

ID	Scenario	Input / steps	Expected result	Actual result	Status
TC-07	Matching claim end to end	Headline that the outlet published	Source Confirmed, headline Matched, everything saved	As expected	Pass
TC-08	Wrong claimed date	Correct headline, wrong date	Headline Matched, date Mismatched	As expected	Pass
TC-09	Changed number	Headline with an altered number	Headline Altered	As expected	Pass
TC-10	All search providers fail	Search clients raise errors	Source Incomplete, not Not Found	As expected	Pass
TC-11	Unknown outlet	Outlet that cannot be resolved (fallback disabled)	Rejected before a submission is created	As expected	Pass
TC-12	Identical claim again	Same claim submitted twice	Saved result reused on the new user's own submission	As expected	Pass
TC-13	Incomplete result not reused	Earlier result was Incomplete	Claim is re-queued, not served from cache	As expected (9 tests)	Pass
TC-14	Body similarity display	Claim with body text	Four scores; an unavailable score is never shown as 0	As expected; see Figure 10.6	Pass
TC-15 (M)	Source not found in UI	Claim the outlet did not publish	Not Found, no Fake verdict, other checks hidden	As expected (Figure 10.7)	Pass

9.5.3 Photo Card and Text & Image Verification

Table 9.5: Test Cases — Photo Card and Text & Image

ID	Scenario	Input / steps	Expected result	Actual result	Status
TC-16	Fast acceptance	Upload a card	202 returned before slow extraction; job finishes without the client	As expected	Pass
TC-17	Gemini unavailable	All Gemini requests fail	Submission fails with the temporary-inconvenience message; no verdict	As expected; also seen in Figure 7.9	Pass
TC-18	No headline or outlet	Card without a readable headline or known outlet	Fails with the “valid headline or recognized outlet” message	As expected	Pass
TC-19	Crash recovery	Worker stops during a job	A fresh worker re-claims the job; Gemini is not called again	As expected	Pass
TC-20	Invalid file type	PDF uploaded as image	Rejected with HTTP 415 before acceptance	As expected	Pass
TC-21	Empty article text	Whitespace body with an image	Rejected with HTTP 422	As expected	Pass
TC-22	Duplicate text and image	Same text and image again	Model not run; earlier prediction reused with similarity scores	As expected	Pass
TC-23	Different image	Same text, different image	Not treated as a duplicate	As expected	Pass

9.5.4 Expert Review, Voting and Escalation

Table 9.6: Test Cases — Expert Review and Voting

ID	Scenario	Input / steps	Expected result	Actual result	Status
TC-24	All rules met	Leader $\geq T$, voters $\geq M$, lead $\geq$ margin	Claim finalized	As expected	Pass
TC-25	Threshold, voters or margin missing	One rule not met	Claim stays open	As expected (3 tests)	Pass
TC-26	Tie	Equal top weights	Never finalizes; escalates at the vote limit	As expected	Pass
TC-27	New expert weight	Expert below N completed reviews	Neutral weight 1	As expected	Pass
TC-28	Own claim and double vote	Expert votes on own claim; votes twice	Both rejected	As expected	Pass
TC-29	Time limit	max_review_hours passed with no votes	Escalated; admin notified once	As expected	Pass
TC-30	Admin rights	Admin opens escalated and normal claims	Decides escalated; view-only otherwise; decision is final	As expected	Pass
TC-31	Short justification	Fewer than 50 characters	Error next to the field; vote not sent	As expected	Pass
TC-32 (M)	Escalated claim end to end	Three split votes, admin decides Real	Final decision Real with Voting Details	As shown in Figure 10.17	Pass

9.5.5 Notifications, Fact Explorer and Extension

Table 9.7: Test Cases — Notifications, Fact Explorer and Extension

ID	Scenario	Input / steps	Expected result	Actual result	Status
TC-33	Personal notifications	Each method × user, expert, admin	Preliminary and final notifications once each	As expected (9 cases)	Pass
TC-34	Guest submitter	Claim without an account	No account notification or email	As expected	Pass
TC-35	SMTP failure	Mail server error	Retry without duplicate notifications	As expected	Pass
TC-36	Explorer filters	Reversed date range	Rejected without sending a search	As expected	Pass
TC-37	Explorer service failure	API error	Error kept separate from “no results”, with retry	As expected	Pass
TC-38	Extension OFF/ON	Turn OFF, then ON	Polling and submissions blocked, then resumed	As expected	Pass
TC-39	Account isolation	Switch accounts in the extension	Other account's activity not shown	As expected	Pass
TC-40	Connection settings	Guest, user and expert try to change URLs	Refused; only a server-confirmed admin can save	As expected	Pass
TC-41	Restricted page	Capture on a Chrome internal page	Selection cannot start; clear message	As expected	Pass

9.6 Observed Performance and Model Results

No load test or formal performance benchmark was run. The backend keeps the measured run time of completed checks. A read-only report from 4 October 2026 (docs1/recent-verification-timings.md) lists eight Text & source checks on earlier pipeline versions (v3.1 and v3.2): their pipeline time ranged from 82 to 153 seconds. These are observations of a few checks, not a benchmark. The recorded accuracy of the multimodal model is given in Section 5.1.2.

9.7 Tests Not Executed or Not Verified

Table 9.8: Tests Not Executed in This Run

Item	Reason	Status
Backend integration tests (tests/integration)	Need a PostgreSQL test fixture; the shared client fixture uses an httpx API removed in httpx 0.28	Not executed
Real-model tests (2)	Require BFG_REAL_MODEL_TESTS=1 and local weights	Skipped
Installed-Chrome checks (live capture, OS notifications, browser restart)	Automated extension tests use simulated Chrome APIs	Manual checklist in extension/TESTING.md; not formally recorded
Accuracy of the source pipeline (NLI, NER, thresholds)	No labelled Bangla evaluation set	Not measured
Load and stress testing	Not planned in this project phase	Not executed

9.8 Risks and Contingencies

Table 9.9: Testing Risks and Contingencies

Risk	Contingency
Gemini quota or network errors stop photo card checks	Retries in batches; clear failure message; no false verdict
A news outlet blocks requests or changes its layout	Playwright fallback, several extractors and Incomplete status instead of Not Found
Model behaviour differs from tests that use fakes	Real-model opt-in tests and the expert review step
Data loss on restart	Durable job queue, heartbeats and idempotent persistence
Too few experts to reach agreement	Configurable limits and administrator escalation

10. User Manual

This chapter explains how to use BanglaFactGuard. Sections 10.1 to 10.7 are for every user, Section 10.8 is for experts and Section 10.9 is for administrators. The website is in English and accepts Bangla text.

10.1 Getting Started

Open the website (for a local setup, <http://localhost:4200>). The home page in Figure 10.1 explains the three ways to check a claim. You can use every verification method without an account. Sign in if you want your claims saved in My Submissions and to receive notifications.

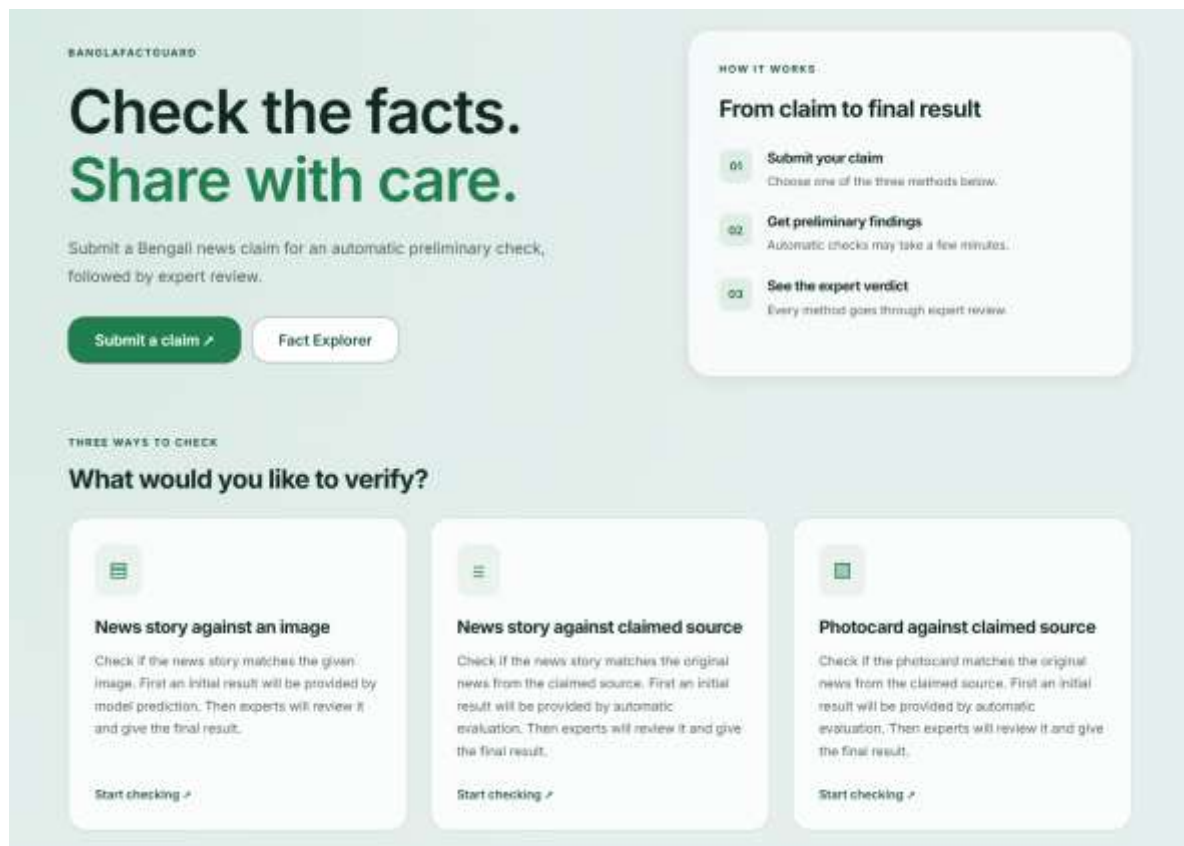


Figure 10.1: Home Page

10.2 Creating an Account and Signing In

1. Click **Get Started** (or **Login**, then **Create one**).
2. Enter your full name, email and a password of at least 8 characters with an uppercase letter and a number. Confirm the password and click **Create Account** (Figure 10.2).
3. To sign in later, click **Login**, enter your email and password and click **Sign In**.
4. If you forget your password, click **Forgot password?**, enter your email, then type the code sent to your email and a new password.

Figure 10.2: Create Account Page

10.3 Checking a Claim

Click **Verify facts** and choose a method card. You can hide the card descriptions with **Hide descriptions**.

10.3.1 Text & Source (News story against claimed source)

1. Type or paste the Bangla headline.
2. Choose the **Claimed news outlet**: the publisher named in the claim, not necessarily the site where you saw it.
3. Optionally paste the article text and choose the claimed publication date.
4. Click **Verify news**. The page shows “Verification in progress” (Figure 7.6). You may leave the page; the check continues.

The form is shown in Figure 7.5. A check usually takes one to two minutes.

10.3.2 Photo Card (Photocard against claimed source)

1. Click the upload box and choose the card image (JPEG, PNG, WebP or GIF, up to 10 MB).
2. Check the preview, then click **Verify photo card**.
3. The system reads the headline, outlet and date from the card. If the card cannot be read, the result page explains why and offers **Try another image** (Figure 7.9).

BanglaFactGuard Home Verify facts Fact Explorer My Submissions About Demo User

Verify facts

Hide descriptions

☐ News story against an image

Check if the news story matches the given image. First an initial result will be provided by model prediction. Then experts will review it and give the final result.

☐ News story against claimed source

Check if the news story matches the original news from the claimed source. First an initial result will be provided by automatic evaluation. Then experts will review it and give the final result.

☒ Photocard against claimed source

Check if the photocard matches the original news from the claimed source. First an initial result will be provided by automatic evaluation. Then experts will review it and give the final result.

Add the photocard

Photocard image *

Click to choose a different image

Selected: photocard.png

Verify photo card

Preview Your Submission

Verified news sources

The system recognizes the following sources as verified Bangla news sources

Show the 10 active outlets

Figure 10.3: Photo Card Form with Preview

10.3.3 Text & Image (News story against an image)

1. Enter the headline and the article text.
2. Upload the news image and check the preview.
3. Click **Check text and image**. The result shows **AI decision: Likely Real** or **Likely Fake**, which is a preliminary estimate only.

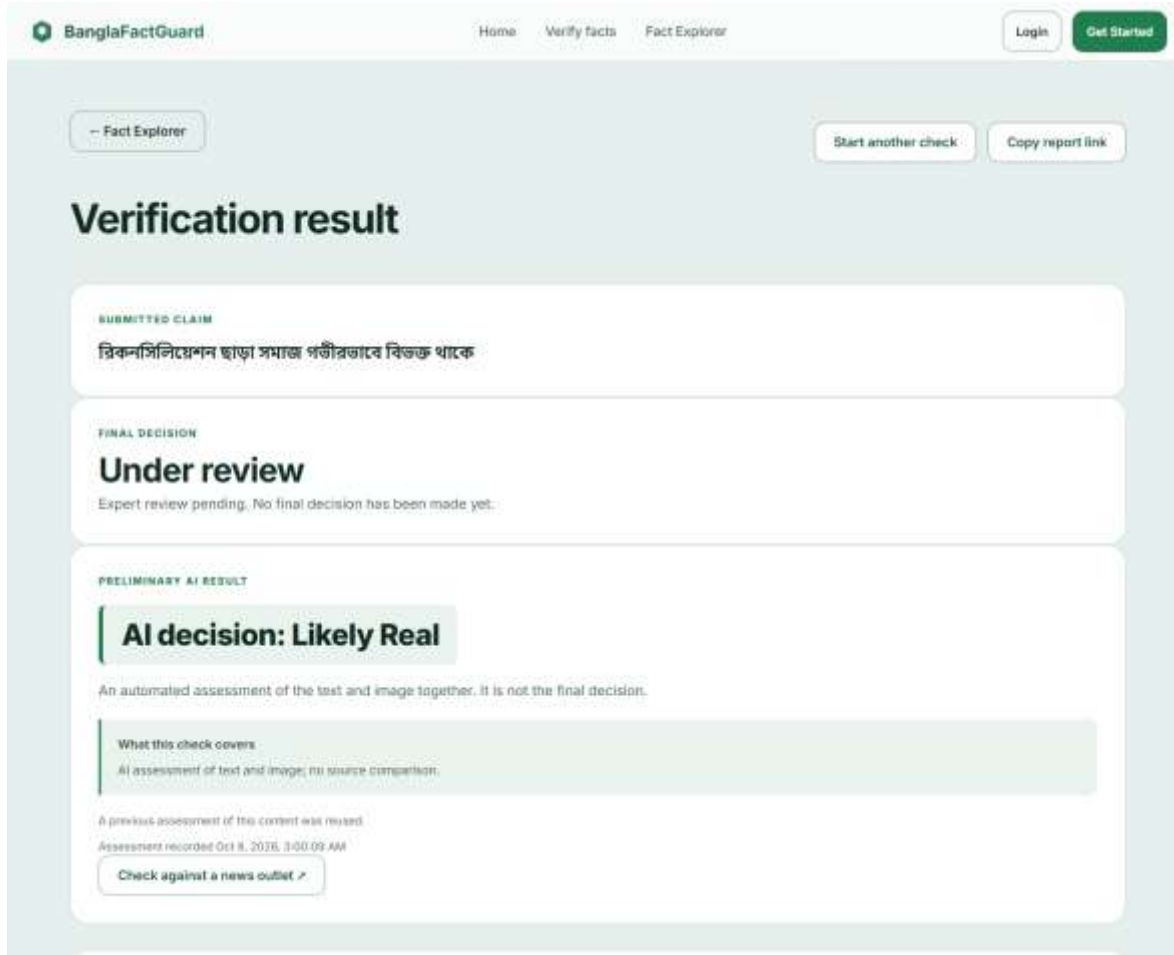


Figure 10.4: Text & Image Result with a Reused Assessment

10.4 Reading a Result

Every result page has the same order: the submitted claim, the **Final decision** box, the **Preliminary result**, the relevant article and the detailed comparisons. While experts are reviewing, the final decision says **Under review**. When a final decision exists, it is shown with **Voting Details**.

- **Relevant article from claimed source:** Found, Not Found or Incomplete. Not Found alone does not mean the news is false.
- **Headline:** Exact Matched, Meaning Preserved or Altered. Altered lists each meaningful difference with the claim text and the source text.
- **Date:** Matched or Mismatched, shown only when you gave a date.
- **Body similarity:** four percentages shown only when you gave article text. They describe overlap, not truth.

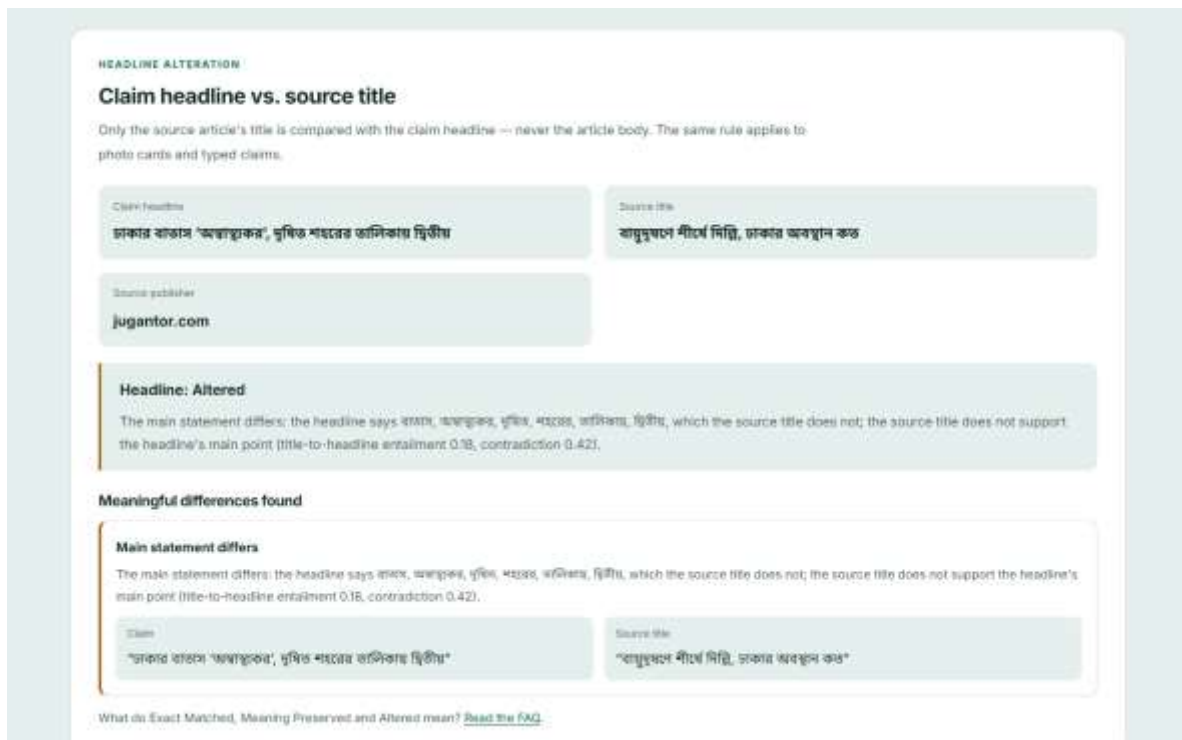


Figure 10.5: Headline Comparison Showing an Altered Headline

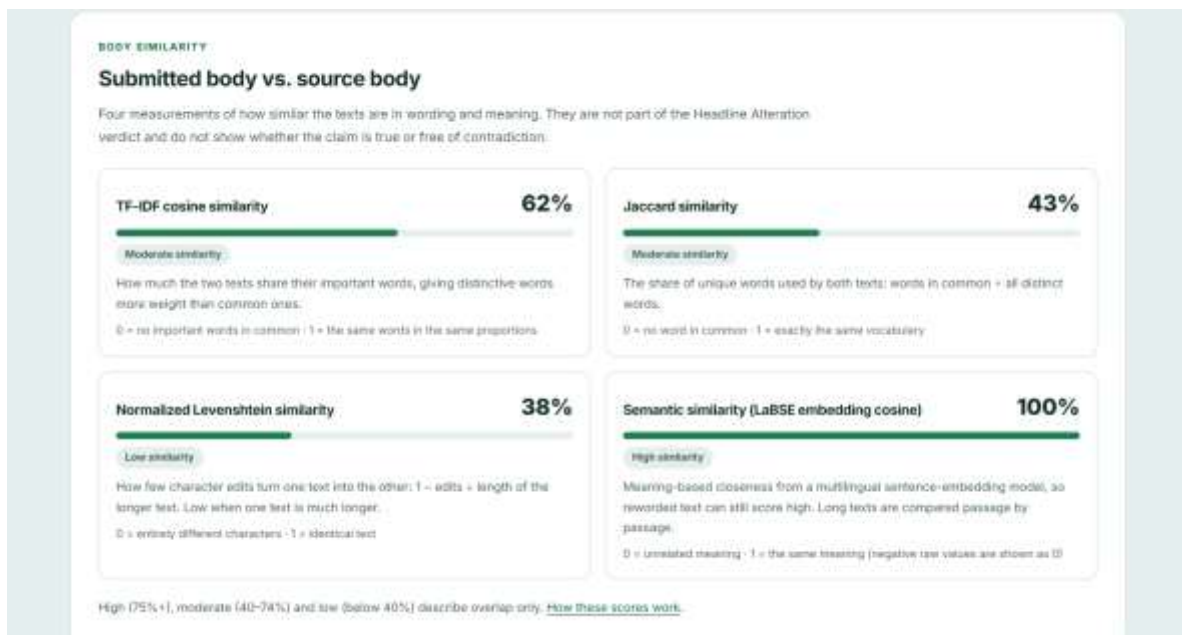


Figure 10.6: Body Similarity Scores

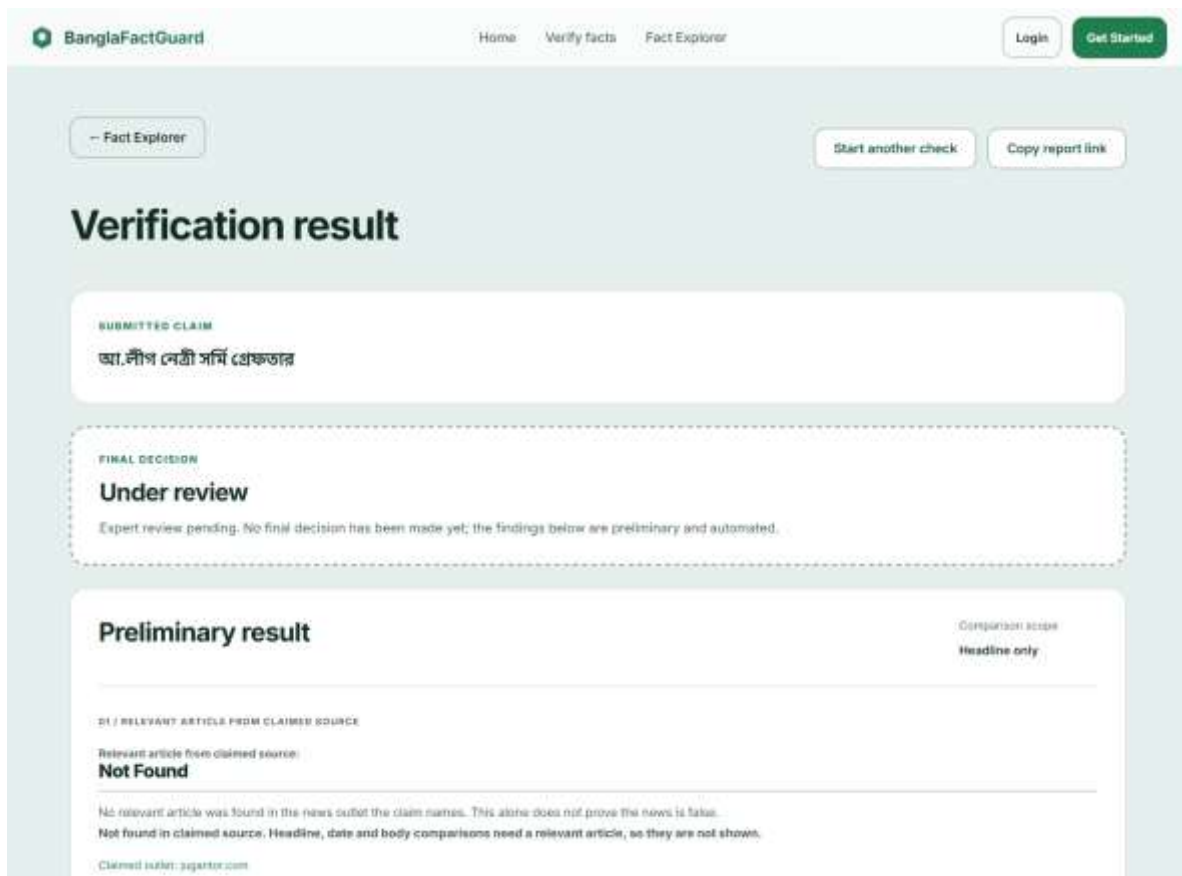


Figure 10.7: Result When the Claimed Source Has No Matching Report

Use **Copy report link** to share a result and **Start another check** to begin again.

10.5 Fact Explorer

Click **Fact Explorer** to search previous claims (Figure 10.8). The top shows archive counts. Choose **Final decisions** or **Under review**, type a keyword, choose a decision and content type and click **Search**. **More filters** adds date and outlet filters. Click **Read full report** to open a claim.

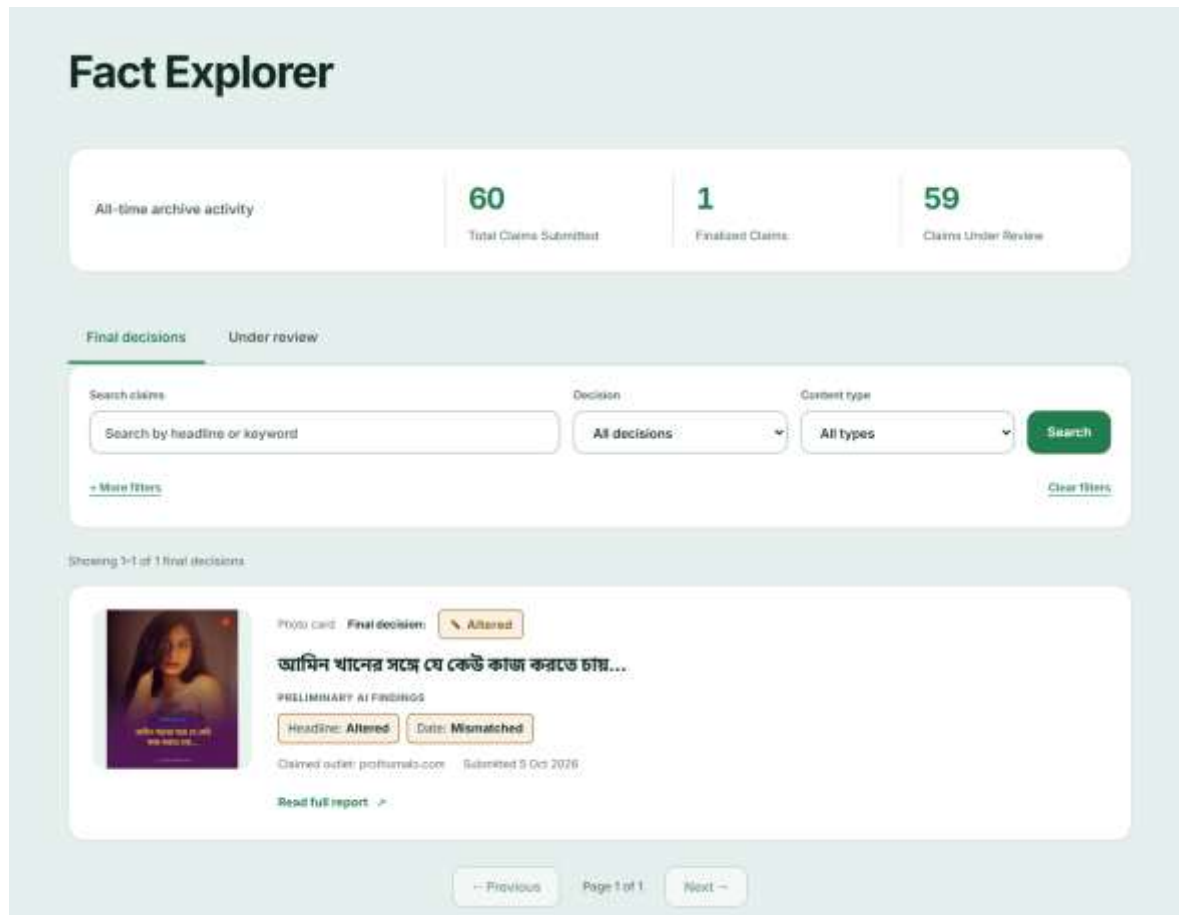


Figure 10.8: Fact Explorer

10.6 My Submissions, Notifications and Settings

Signed-in users can open **My Submissions** to see every claim they submitted, with its method, preliminary findings and review status (Figure 10.9). The bell icon opens **Notifications** (Figure 10.10). Each notification has **View result** and **Mark as read**; **Mark all as read** clears the list.

Final results are also sent by email when email is configured. **Account settings** (user menu) changes your name and password.

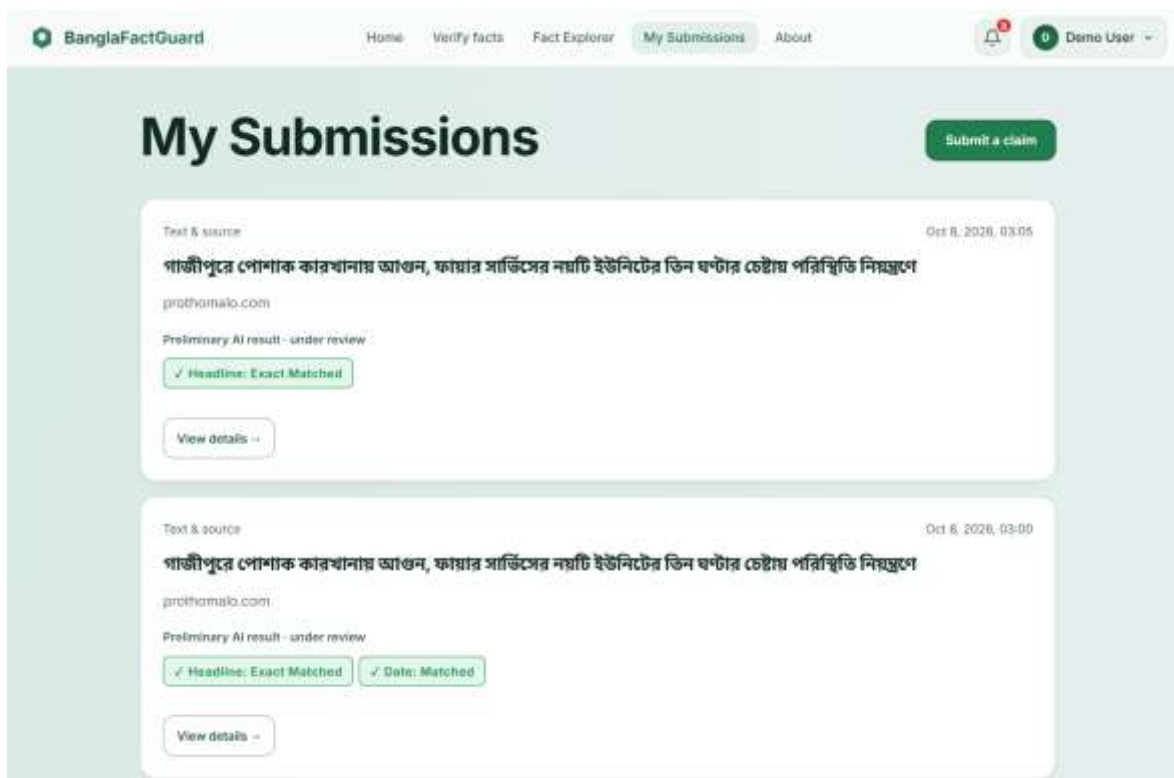


Figure 10.9: My Submissions

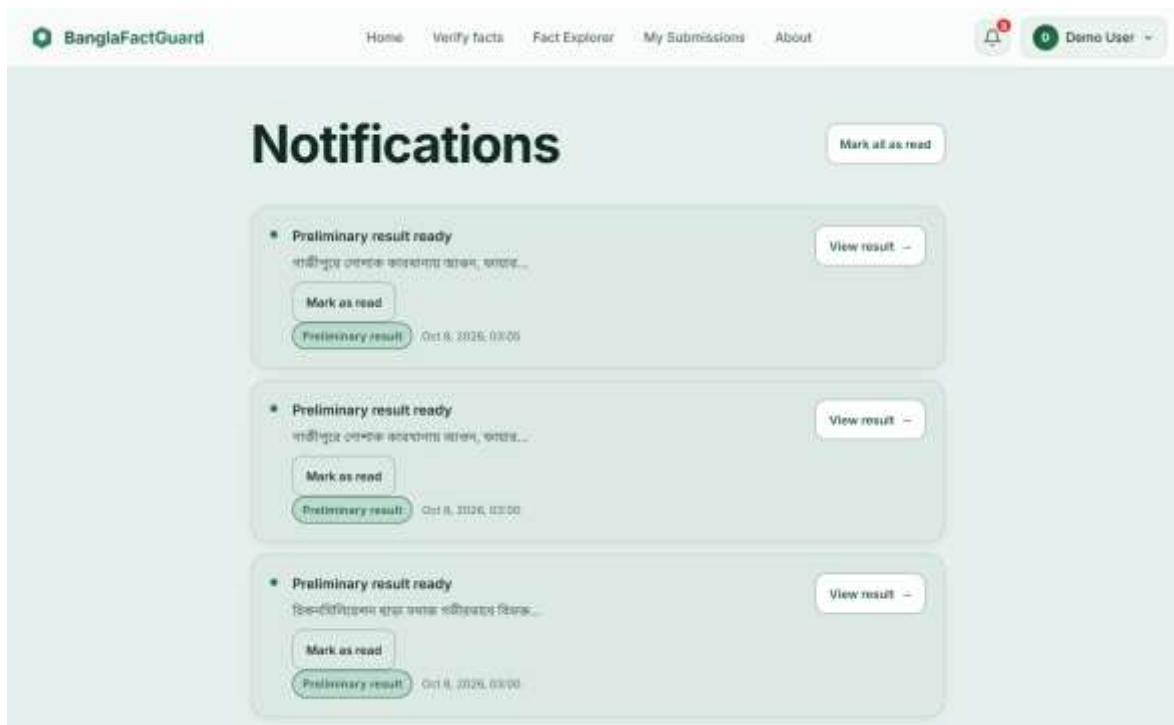


Figure 10.10: Notifications

10.7 Using the Browser Extension

1. Pin **BanglaFactGuard — Browser Extension** in Chrome and click its toolbar icon on the news page. The side panel opens.
2. Choose **Text & source**, **Photo card** or **Text & image**.
3. Select text on the page and click **Use selected text** or type it. For images, click **Upload image** or **Select screenshot area** and drag over the part of the page you want.
4. Adjust the image with **Apply crop**, **Reset crop** or **Remove**.
5. Click **Submit for verification**. The **Activity** tab shows “Checking your claim” and later “Preliminary result ready” with a desktop alert. Click **View details** to open the full report on the website.
6. Use the **Account** tab to sign in with your website account and the ON switch to pause the extension.

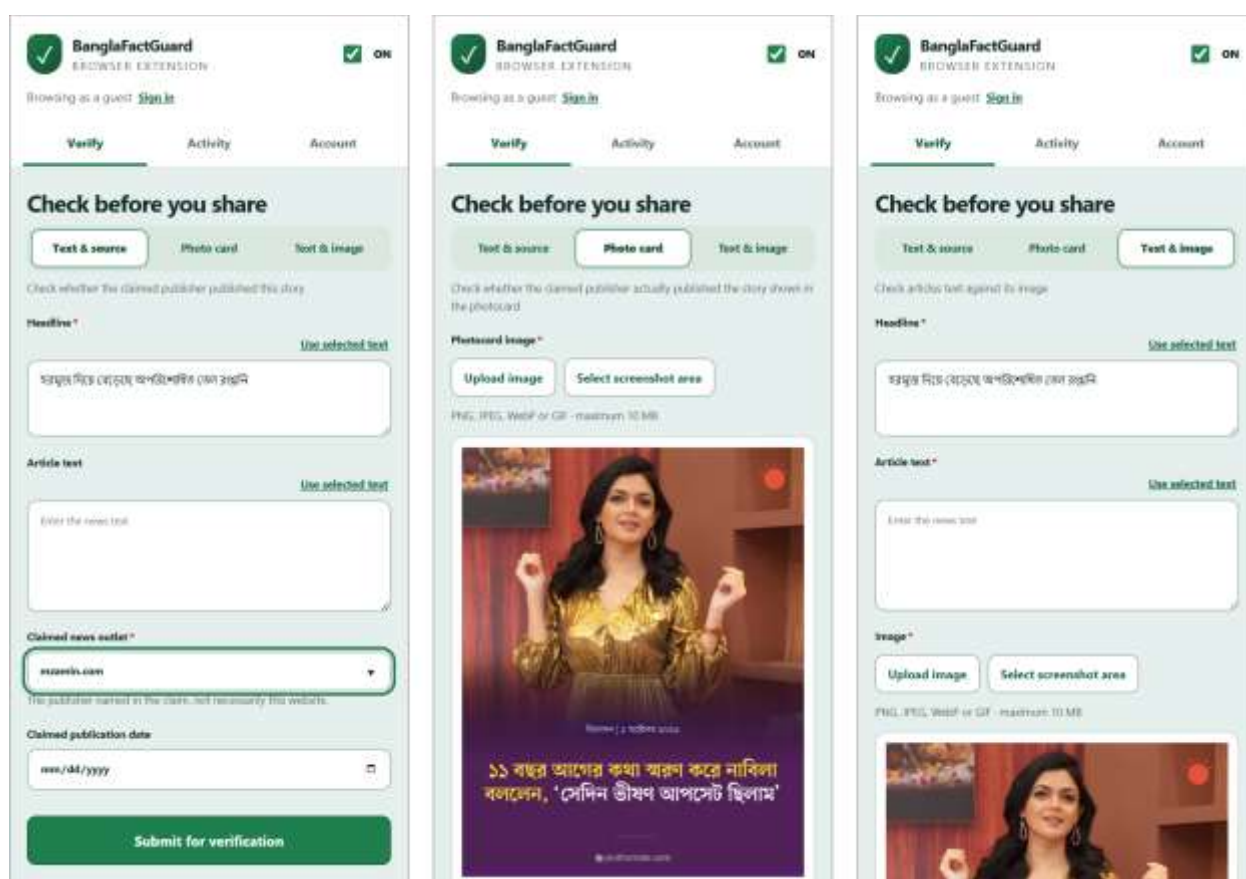


Figure 10.11: Extension Verify Tab — Text & Source, Photo Card and Text & Image

The account and activity screens are shown in Figure 7.10.

10.8 Expert Tasks

1. Sign in with the account created by the administrator. Open **Expert Queue** (Figure 10.12).
2. Search if needed and click **Review & vote** on a claim.

3. Read the claim, the preliminary findings and the article (Figure 7.11). Remember that automated findings are supporting information, not instructions.
4. Choose **Real**, **Fake**, **Misleading** or **Altered**, optionally add supplementary findings, write a justification of at least 50 characters and click **Submit vote**.
5. Open **Review History** to see your votes and the final decisions. **Account settings** shows your number of reviews; credibility weight activates after the required number of reviews.

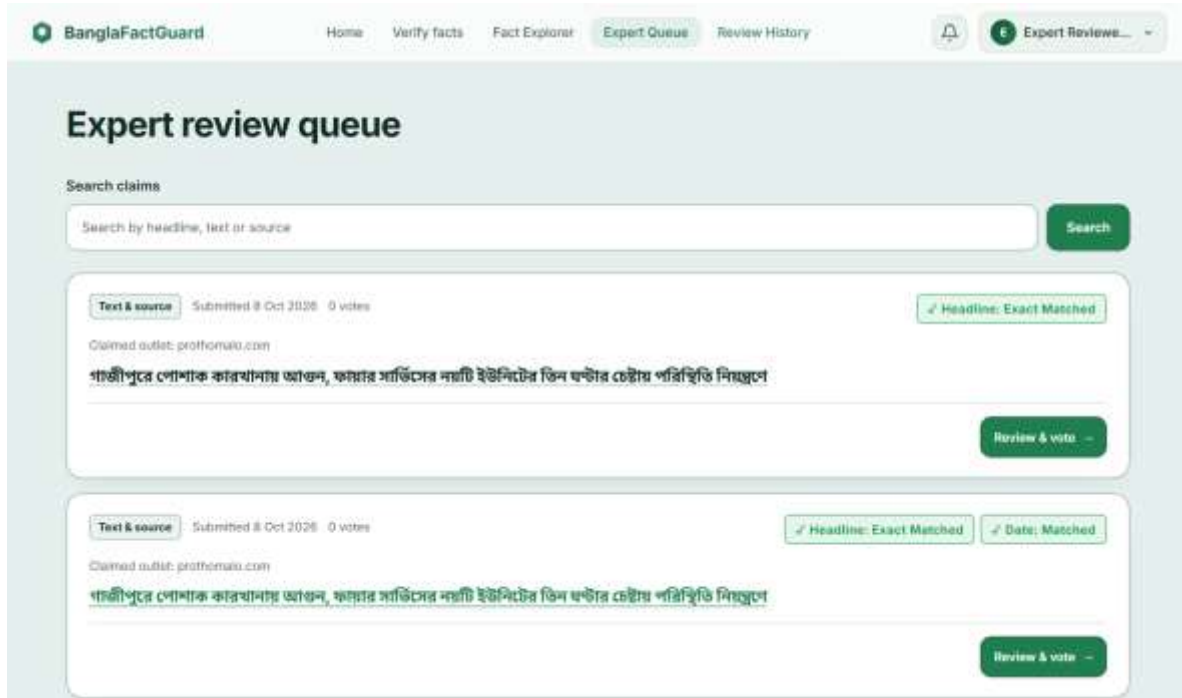


Figure 10.12: Expert Review Queue

10.9 Administrator Tasks

The **Admin dashboard** (Figure 10.13) shows escalated claims that need your decision, activity counts, claims waiting longest and recent decisions. Use the **Experts**, **Sources** and **Voting & tiers** buttons to manage the system.

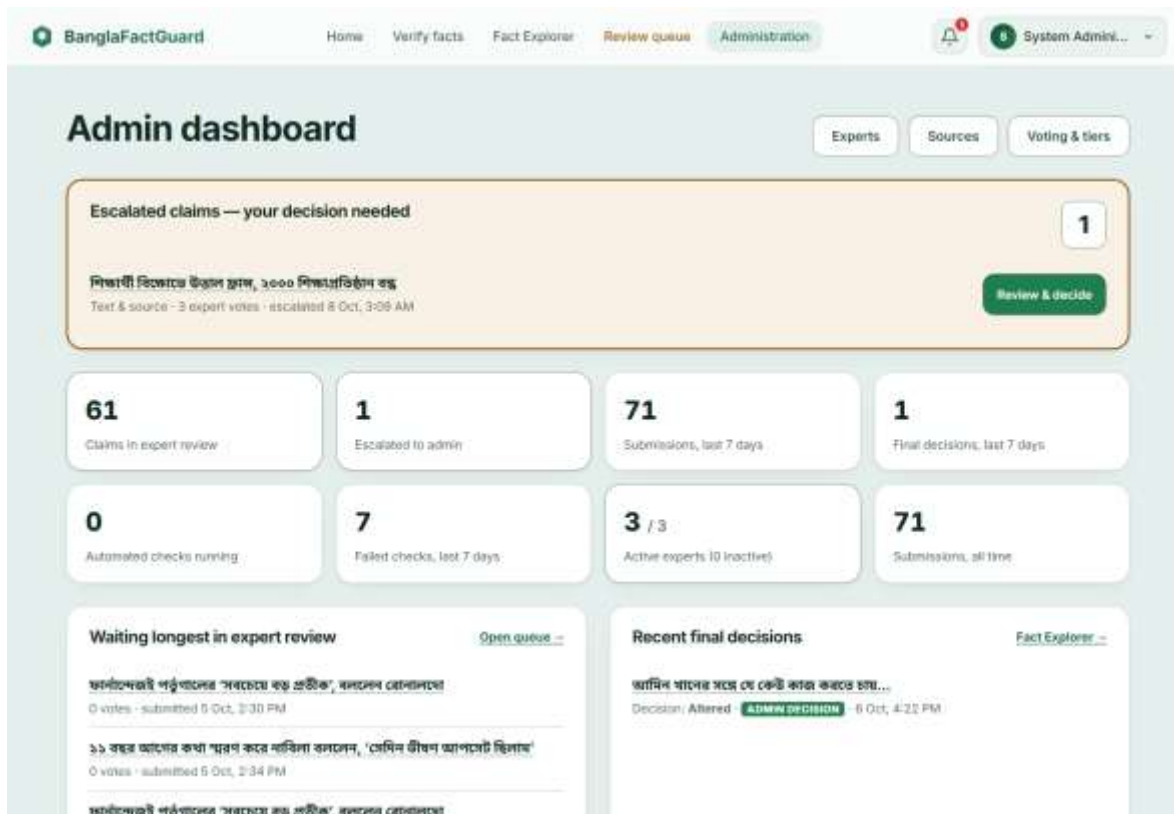


Figure 10.13: Administrator Dashboard

- **Experts:** click **Create expert**, enter name, email, password and expertise area and click **Create Expert Account**. Use **Edit**, **Deactivate** and **Reset password** in the table (Figure 10.14).
- **Sources:** click **Register New Source** to add an outlet or use **Edit**, **Disable** and **Delete** on a card (Figure 10.15). Only active sources are used for verification.
- **Voting & tiers:** set Min voters (M), Verified threshold (T), Lead margin, Activation threshold (N), Max review votes and Max review hours, then **Save Configuration**. Add or edit tiers with their accuracy range and weight (Figure 10.16).
- **Escalated claims:** in **Review queue**, choose **Escalated — needs your decision**, click **Review & decide**, choose the verdict, explain it and submit. Your decision is final; the public result then shows it with Voting Details (Figure 10.17). Claims that are not escalated are view-only.

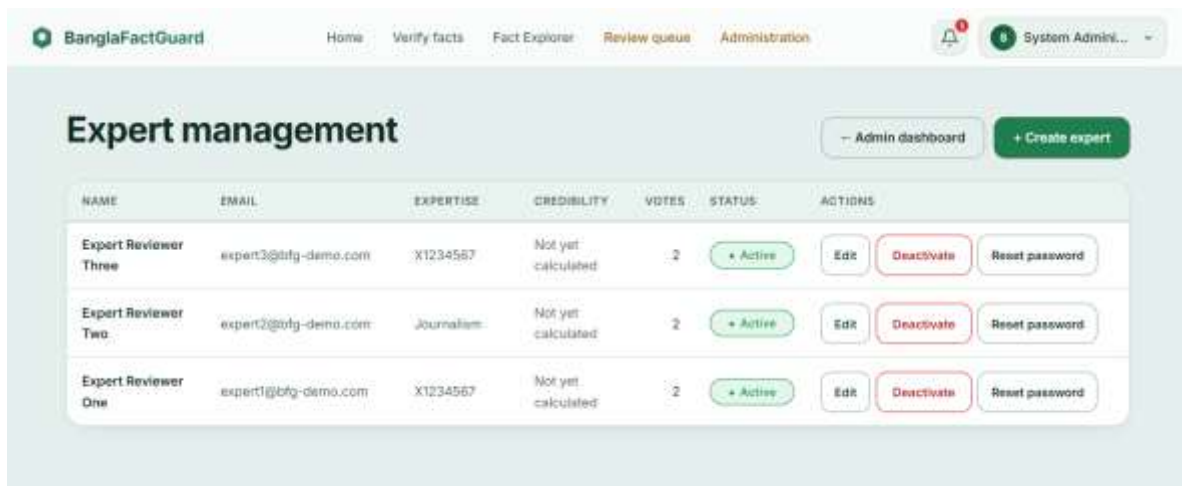


Figure 10.14: Expert Management

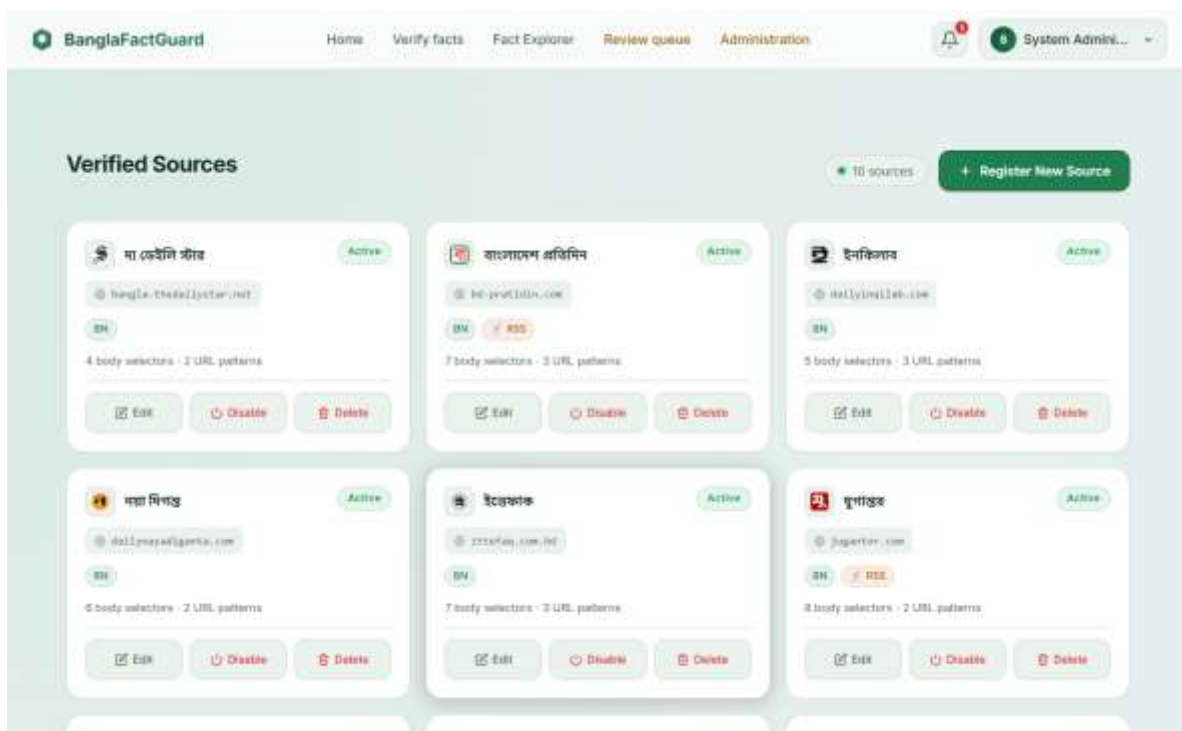


Figure 10.15: Verified Sources

BanglaFactGuard
Home
Verify facts
Fact Explorer
Review queue
Administration
System Admin...

Credibility Weight Tiers

Back to Admin
New Tier

Voting Configuration

Min voters (M)

Verified threshold (T)

Lead margin

Activation threshold (N)

Max review votes

3

2

1

10

3

Max review hours

Not configured

Save Configuration

Label	Accuracy Range	Weight	Status	
Novice	0% - 40%	+0.5	Active	Edit Delete
Competent	40% - 70%	+1	Active	Edit Delete
Expert	70% - 90%	+1.5	Active	Edit Delete
Master	90% - 100%	+2	Active	Edit Delete

Figure 10.16: Voting Configuration and Credibility Weight Tiers

BanglaFactGuard
Home
Verify facts
Fact Explorer
Review queue
Administration
System Admini...

Fact Explorer
Start another check
Copy report link

Verification result

SUBMITTED CLAIM

শিক্ষার্থী বিক্ষোভে উদ্ভাল ফ্রান্স, ২০০০ শিক্ষাপ্রতিষ্ঠান বন্ধ

FINAL DECISION

Final decision ✓ Real

Expert reviewers could not agree, so an administrator made this final decision. The preliminary AI findings below are kept for reference; they are not the final decision.

+ Voting Details

Expert Reviewer One Expert

I compared the claim headline with the source article title and the full report. The details shown support this verdict.

Voted 8 Oct 2026, 3:09 AM

Vote: ✓ Real

Expert Reviewer Two Expert

I compared the claim headline with the source article title and the full report. The details shown support this verdict.

Voted 8 Oct 2026, 3:09 AM

Vote: X Fake

Expert Reviewer Three Expert

I compared the claim headline with the source article title and the full report. The details shown support this verdict.

Voted 8 Oct 2026, 3:09 AM

Vote: ⚠ Misleading

System Administrator Admin FINAL DECISION

I compared the claim headline with the source article title and the full report. The details shown support this verdict.

Voted 8 Oct 2026, 3:10 AM

Vote: ✓ Real

Figure 10.17: Final Decision with Voting Details After an Administrator Decision

10.10 Troubleshooting

Table 10.1: Common Problems and Solutions

Problem	What to do
“Unknown claimed source” or the outlet is missing	Choose the publisher named in the claim; ask an administrator to register or activate the outlet.
The claim stays “in progress” for a long time	Checks may take a few minutes. The backend must be running with background jobs enabled.
Photo card “temporarily unavailable”	Gemini could not be reached; submit again later.
Photo card has no valid headline or outlet	Use a clearer card that shows the headline and the outlet's name or logo.
Extension cannot capture a page	Click the pinned toolbar icon on that page first; Chrome internal pages cannot be captured, so upload an image instead.
No desktop alert	Check that the extension is ON, notifications are enabled in the Account tab and Chrome may show notifications; Activity still shows the result.

11. Repository Installation and Deployment

11.1 Repository

The source code is publicly available at <https://github.com/abulkalamazad1425/BanglaFactGuard>. Clone it with:

```
git clone https://github.com/abulkalamazad1425/BanglaFactGuard.git
cd BanglaFactGuard
```

11.2 Prerequisites

Table 11.1: Prerequisites

Software	Version / note
Python	3.11 or newer, with PyTorch and model dependencies
Node.js and npm	Node 22 for the website and the extension
PostgreSQL	A database and user for the project (PostgreSQL 13+; 17 used in development)
Docker Compose	For the included Redis and MinIO services
Google Chrome	Desktop Chrome 120 or newer for the extension
Gemini API key	Needed for photo card reading
Multimodal model files	img_backbone.pt, text_backbone.pt, classifier.pt and tokenizer/ in MULTIMODAL_MODEL_DIR

11.3 Backend Setup

Run these PowerShell commands from the repository root:

```
cd backend
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m playwright install chromium
Copy-Item .env.example .env
```

Edit backend/.env with your own values (Table 11.2). Generate a secret key with `python -c "import secrets; print(secrets.token_urlsafe(48))"`. Never commit the .env file.

Table 11.2: Main Configuration Settings (backend/.env)

Setting	Purpose
DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD	PostgreSQL connection
REDIS_HOST, REDIS_PORT	Redis connection
MINIO_*	Image storage endpoint, credentials and bucket
AUTH_SECRET_KEY	Required JWT signing key (at least 32 characters)
ML_*	LaBSE, NLI and NER model names (ML_NER_MODEL_NAME=arafatfahim/BanglaTag)
GEMINI_API_KEY, GEMINI_*	Photo card reading and request batches
MULTIMODAL_MODEL_DIR, MULTIMODAL_LOAD_ON_STARTUP	Text & image model files and loading
JOBS_ENABLED, JOBS_*	Background worker and recovery
EMAIL_SMTP_*, EMAIL_FROM_ADDRESS, EMAIL_WEBSITE_URL	OTP and final-result emails

11.4 Services, Database and Sources

Create the PostgreSQL database named in DB_NAME. Then, from backend/, start Redis and MinIO, apply the migrations, seed the verified sources and start the API:

```
docker compose up -d
.\.venv\Scripts\python.exe -m alembic upgrade head
.\.venv\Scripts\python.exe scripts/seed_verified_sources.py
.\.venv\Scripts\python.exe run.py
```

The API runs at <http://localhost:8000> and its interactive documentation at <http://localhost:8000/docs>. The migrations include an initial administrator account; experts are created by the administrator.

11.5 Website

```
cd frontend
npm ci
npm start
```

Open <http://localhost:4200>. The development API address is set in frontend/src/environments/environment.ts (default <http://localhost:8000/api/v1>). A production build is made with `npm run build` and needs a web server that forwards /api/v1 to FastAPI.

11.6 Browser Extension (Installation)

```
cd extension
npm ci
npm run build
```

1. Open <chrome://extensions> and turn on **Developer mode**.
2. Click **Load unpacked** and select the extension/dist folder.
3. Pin **BanglaFactGuard — Browser Extension** and click its icon on a news page.

11.7 Running the Tests

```
cd backend; .\.venv\Scripts\python.exe -m pytest tests/unit -q --no-cov
cd frontend; npx ng test --watch=false --browsers=ChromeHeadless
cd extension; npm test
```

11.8 Deployment Status

- **Verified:** the local deployment in Figure 6.6 (Angular dev server, uvicorn API, PostgreSQL 17 on the host, Redis and MinIO in Docker, unpacked extension) was used for the demonstration and the screenshots in this report.
- **Documented but not verified:** a production deployment with the built website behind a web server that proxies /api/v1. No public hosting was set up in this project.

12. Conclusion

This report presented BanglaFactGuard, a multimodal fact-checking system for Bangla news. The final system lets anyone check a claim in three ways from the website or the browser extension: against the claimed news outlet, from a photocard or as a text and image pair. Automated findings are fast and always marked as preliminary, while the final verdict comes from credibility-weighted expert voting, with administrator decisions when experts do not agree.

Compared with the SRS, several designs changed during implementation. The single automated True/False/Partially True label was replaced by separate Source, Headline and Date findings; OCR was replaced by Gemini reading of photocards; and background jobs were moved to a durable database queue. These changes make the results easier to understand and the system more reliable. All 550 executed automated tests passed.

The system still has limitations. The NLI and NER models have not been evaluated on labelled Bangla data, the multimodal model reached 80% recorded test accuracy, photocard reading depends on the Gemini service and the integration tests and a production deployment were not completed. Future work includes a labelled Bangla evaluation set, more verified outlets, load testing, a public deployment and support for video claims.

13. References

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